

AKADEMIKA

**CYBER SECURITY
ESSENTIALS**

EXPERIMENTAL MANUAL

.... A MESSAGE FROM

Today's system designers are faced with tomorrow's problems. NETWORKING is one of the important subjects which need to be taught while learning electronics.

There are many benefits of networking. "Resource sharing- using network- connected peripheral devices like printers, scanners and copiers, or sharing software between multiple users, saves money. Sharing a single internet connection- it is cost efficient and can help protect your systems if you properly secure the network".

OUR MOTTO;

-Light years ahead—refers to leadership.

As leaders in our industry in India, we are totally committed to servicing as the standard against which all are measured in the areas of

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We are truly light years ahead of our competition in this area. That means that youour valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have completely truly innovative & superior training products. We are so committed toquality that we back our products with a complete comprehensive

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SAFETY RULES

- Read carefully and follow the instructions mentioned in this manual. This user manual includes all the important points about the installation, use and the maintenance of the product. Keep this manual always with you, for quick reference.
- After unpacking the product, arrange all the accessories in proper order, so that their integrity is checked with the packing list. Also, ensure that the accessories have no visible damage.
- Before connecting the power supply to the kit, be sure that the jumpers and the connecting chords are connected correctly, as per the experiment.
- This kit must be employed only for the use for which it has been conceived, i.e. as educational kit and must be used under the direct survey of expert personnel. Any other use is inadvisable and dangerous too. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.
- In case of any fault or malfunctioning in the trainer kit, turn off the power supply. Please do not tamper the kit. If you require our service, kindly contact the service centre for technical assistance.
- The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we would charge for repairs, regardless of the warranty period. The warranty does not cover include perishable items like connecting chords, crystals, etc. and other imported items.

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The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product will be Akademika's sole judgment. The defective product will be replaced with a new one or repaired, without charge or with charge.

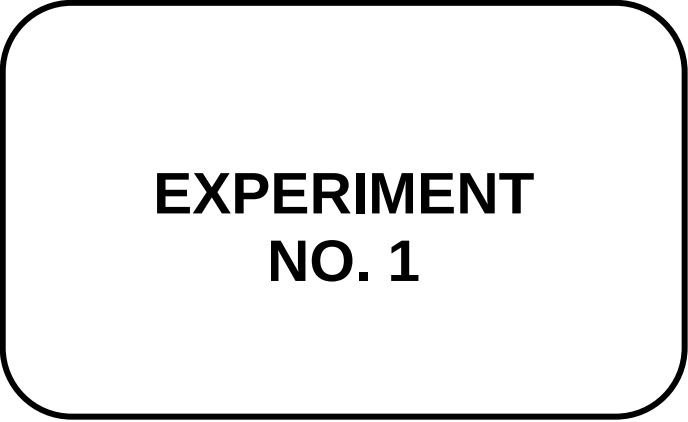
In the warranty period if the service is needed, the purchaser should get in touch with the service center or the sales outlet. The purchaser should return the product to the service center or the sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of the kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case the warranty does not cover the product on Akademika's judgment, we would repair the product after obtaining prior permission from purchaser who would receive an estimate statement about the repairing charges. In such cases, Akademika's bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but they have no rights to stretch the meaning of original warranty contents or to offer an additional warranty. Akademika's does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika's is not responsible for the damages even if it is notified of above dangers in advance as well.

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EXPERIMENT NO. 1

EXPERIMENT NO.1

NAME:-

Create and Store Strong Passwords.

OBJECTIVE:-

Understand the concepts behind a strong password.

Part 1: Explore the concepts behind creating a strong password.

Part 2: Explore the concepts behind securely storing your passwords?

BACKGROUND / SCENARIO:-

Passwords are widely used to enforce access to resources. Attackers will use many techniques to learn users' passwords and gain unauthorized access to a resource or data. To better protect yourself, it is important to understand what makes a strong password and how to store it securely.

REQUIRED RESOURCES:-

- PC or mobile device with Internet access

Part 1: Creating a Strong Password: -

Strong passwords have four main requirements listed in order of importance:

1. The user can easily remember the password.
2. It is not trivial for any other person to guess a password.
3. It is not trivial for a program to guess or discover a password.
4. Must be complex, containing numbers, symbols and a mix of upper case and lowercase letters.

Based on the list above, the first requirement is probably the most important because you need to be able to remember your password. For example, the password **#4ssFrX^aartPOknx25_70!xAdk<d!** is considered a strong password because it satisfies the last three requirements, but it is very difficult to remember.

Many organizations require passwords to contain a combination of numbers, symbols, and lower and uppercase letters. Passwords that conform to that policy are fine as long as they are easy for the user to remember. Below is a sample password policy set for a typical organization

- The password must be at least 8 characters long.
- The password must contain upper- and lower-case letters.
- The password must contain a number.
- The password must contain a non-alphanumeric character.

Take a moment to analyze the characteristics of a strong password and the common password policy set shown above. Why does the policy set neglect the first two items? Explain.

Adding symbols, numbers and mixed upper/lower case to a password makes it harder for the user to remember it. Traditionally, when a user finds a password that conforms to a specific set of re-use the same structure or even the entire password through other services. Some systems will also force the user to change the password periodically, keeping users from using past passwords again. Those users are also very likely to introduce minor changes to the password instead of creating an entirely different password that still conforms to the given password policies.

A good way to create strong passwords is to choose four or more random words and string them together. The password **televisionfrogbootschurch** is stronger than **J0n@than#81**. Notice that while the second password is in compliance with the policies described above, password cracker programs are very efficient at guessing that type of password. While many password policy sets will not accept the first password, **televisionfrogbootschurch**, it is much stronger than the second. It is easier for the user to remember (especially is associated with an image), it is very long and its random factor makes it hard for password crackers to guess it.

Using an online password creation tool, create passwords based on the common company password policy set described above.

- a) Open a web browser and go to <http://passwordsgenerator.net>
- b) Select the options to conform to password policy set
- c) Generate the password.

Is the password generated easy to remember?

Answers will vary. But it is very likely the password will not be easy to remember.

Using an online password creation tool, create passwords based on random words. Notice that because the words are appended together, they are not seen as dictionary words

- d) Open a web browser and go to <http://preshing.com/20110811/xkcd-password-generator/>
- e) Generate a random word password by clicking **Generate Another!** at the top portion of the webpage.
- f) Is the password generated easy to remember?

Answer will vary. But it is very likely the password will be easy to remember.

Part 2: Securely Storing Passwords

If the user chooses to use a password manager, the first strong password characteristic can be dropped because the user has access to the password manager at all times. Notice that some users only trust their passwords to their own memory. Password managers, either local or remote, must have a password store, and it can be compromised.

The password manager password store must be strongly encrypted and access to it must be tightly controlled. With mobile phone apps and web interfaces, cloud-based password managers provide anytime, uninterrupted access to its users.

A popular password manager is Last Pass. Create a trial Lastpass account:

- a) Open a web browser and go to <https://lastpass.com/>
- b) Click **Start Trial** to create a trial account.
- c) Fill out the fields, as instructed.
- d) Set a master password. This password gives you access to your LastPass account.
- e) Download and install the LastPass' client for your operating system.
- f) Open the client and log in with your LastPass master password.
- g) Explore LastPass password manager.

As you add passwords to Lastpass, where are the passwords stored?

The passwords are stored on the cloud, on Lastpass' servers.

Besides you, at least one other entity has access to your passwords. Who is that entity?

Lastpass

While having all your passwords stored on the same place can be convenient, there are drawbacks. Can you think of any?

Answers will vary. Lastpass' servers become a big target for attackers as it contains many users' passwords. The responsibility of maintaining your passwords were now delegated to a third party company which you have no control on their security policies. You choose to trust they are doing a good job at protecting your passwords but there's no guarantees.

Part 3 : What Is a Strong Password Then?

Using on the strong password characteristics given at the beginning of this lab, choose a password that is easy to remember but hard to be guessed. Complex passwords are OK as long as it does not impact more important requirements such as the ability to easily remember it.

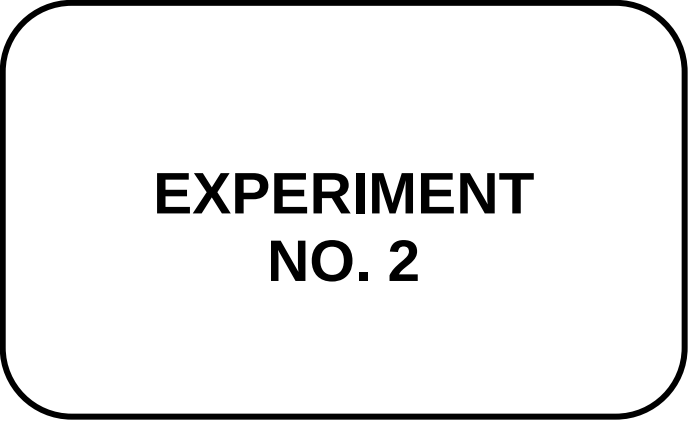
If a password manager is used, the need to be easily remembered can be relaxed. Below is a quick summary:

Choose a password you can remember.

Choose a password that someone else cannot associate with you.

Choose different passwords and never use the same password for different services.

Complex passwords are OK as long as it does not become harder to remember.



EXPERIMENT NO. 2

EXPERIMENT NO. 2

NAME:-

Discover Your Own Risky Online Behaviour.

OBJECTIVES:-

Explore actions performed online that may compromise your safety or privacy.

BACKGROUND / SCENARIO:-

The Internet is a hostile environment, and you must be vigilant to ensure your data is not compromised. Attackers are creative and will attempt many different techniques to trick users. This lab helps you identify riskyonline behavior and provide tips on how to become safer online.

Part 1: Explore the Terms of Service Policy

Answer the questions below with honesty and take note of how many points each answer gives you. Add allpoints to a total score and move on to Part 2 for an analysis of your online behavior.

What kind of information do you share with social media sites? _____

Everything; I rely on social media to keep in touch with friends and family.

(3 points)Articles and news I find or read (2 points)

It depends; I filter out what I share and with whom I share. (1 point)

Nothing; I do not use social media. (0 points)

When you create a new account in an online service, you: _____

Re-use the same password used in other services to make it easier to remember.

(3 points)Create a password that is as easy as possible so you can remember it.

(3 points) Create a very complex password and store it in a password manager service.

(1 point) Create a new password that is similar to, but different from, a password used in another service. (1point)

Create an entirely new strong password.

(0 points)When you receive an email with links to other sites: _____

You do not click the link because you never follow links sent to you via email. (0 points)

You click the links because the email server has already scanned the email. (3 points)

You click all links if the email came from a person you know. (2 points)

You hover the mouse on links to verify the destination URL before clicking. (1 point)

A pop-up window is displayed as you visit a website. It states your computer is at risk and you shoulddownload and install a diagnostics program to make it safe:_____

You click, download, and install the program to keep your computer safe. (3 points) You inspect the pop-up windows and hover over the link to verify its validity. (3 points) Ignore the message, making sure you don't click it or download the program and close the website. (0 points)

When you need to log into your financial institution's website to perform a task, you:

Enter your login information immediately. (3 points)

You verify the URL to ensure it is the institution you were looking for before entering any information. (0 points) You don't use online banking or any online financial services. (0 points)

You read about a program and decide to give it a try. You look around the Internet and find a trial version on an unknown site, you: _____

Promptly download and install the program. (3 points)
Search for more information about the program creator before downloading it. (1 points) Do not download or install the program. (0 points)
You find a USB drive while walking to work. you: _____

Pick it up and plug it into your computer to look at its contents. (3 points)
Pick it up and plug it into your computer to completely erase its contents before re-using it. (3 points)
Pick it up and plug it into your computer to run an anti-virus scan before re-using it for your own files (3 points)
Don't pick it up. (0 points)
You need to connect to the Internet and you find an open Wi-Fi hotspot. _____

You: Connect to it and use the Internet. (3 points)
Don't connect to it and wait until you have a trusted connection. (0 points)
Connect to it and establishes a VPN to a trusted server before sending any information. (0 points)

Part 2: Analyze Your Online Behavior

The higher your score, the less safe your online behaviors are. The goal is to be 100% safe by paying attention to all your online interactions. This is very important as it only takes one mistake to compromise your computer and data.
Add up the points from Part 1. Record your score. _____

0: You are very safe online.

0 – 3: You are somewhat safe online but should still change your behavior to be completely safe.

3 – 17: You have unsafe behavior online and have a high risk of becoming compromised.

18 or more: You are very unsafe online and will be compromised. Below are a few important online safety tips.

The more information you share on social media, the more you allow an attacker to know you. With more knowledge, an attacker can craft a much more targeted attack. For example, by sharing with the world you went to a car race, an attacker can craft a malicious email coming from the ticketing company responsible for the race event. Because you have just been to the event, the email seems more credible.

Reusing passwords is a bad practice. If you reuse a password in a service under attackers' control, they may be successful when attempting to log in as you in other services.

Emails can be easily forged to look legitimate. Forged emails often contain links to malicious sites or malware. As a general rule, do not click embedded links received via email.

Do not accept any unsolicited software, especially if it comes from a web page. It is extremely unlikely that a web page will have a legitimate software update for you. It is strongly recommended to close the browser and use the operating system tools to check for the updates.

Malicious web pages can be easily made to look like a bank or financial institution website. Before clicking the links or providing any information, double-check the URL to make sure it is the correct web page.

When you allow a program to run on your computer, you give it a lot of power. Choose wisely before allowing a program to run. Research to make sure the company or individual behind the program is a serious and legitimate author. Also, only download the program from the official website of the company or individual.

USB drives and thumb drives include a tiny controller to allow computers to communicate with it. It is possible to infect that controller and instruct it to install malicious software on the host computer. Because the malware is hosted in the USB controller itself and not in the data area, no amount of erasing or anti-virus scanning will detect the malware.

Attackers will often deploy fake Wi-Fi hotspots to lure users. Because the attacker has access to all the information exchanged via the compromised hotspot, users connected to that hotspot are at risk. Never use unknown Wi-Fi hotspots without encrypting your traffic through a VPN. Never provide sensitive data such as credit card numbers while using an unknown network (wired or wireless).

Reflection:-

After analyzing your online behavior, what changes would you make to protect yourself online?

Answers will vary.

**EXPERIMENT
NO. 3**

EXPERIMENT NO.3

NAME:-

Using Digital Signature.

OBJECTIVE:-

Understand the concepts behind digital signature.

Part 1: Demonstrate the use of digital signatures.

Part 2: Demonstrate the verification of a digital signature.

BACKGROUND/SCENARIO

A digital signature is a mathematical technique used to validate the authenticity and integrity of a digital message. A digital signature is the equivalent of a handwritten signature. Digital signatures can actually be far more secure. The purpose of a digital signature is to prevent the tampering and impersonation in digital communications. In many countries, including the United States, digital signatures have the same legal significance as traditional forms of signed documents. The United States Government now publishes electronic versions of budgets, laws, and congressional bills with digital signatures

REQUIRED RESOURCES: -

PC or mobile device with Internet access

Part 3: Using Digital Signatures

In this part, you will use a website to verify a document signature between Alice and Bob. Alice and Bob share a pair of private and public RSA keys. Each of them uses their private key to sign a legal document. They then send the documents to each other. Both Alice and Bob can verify each other's signature with the public key.

They must also agree on a shared public exponent for calculation.

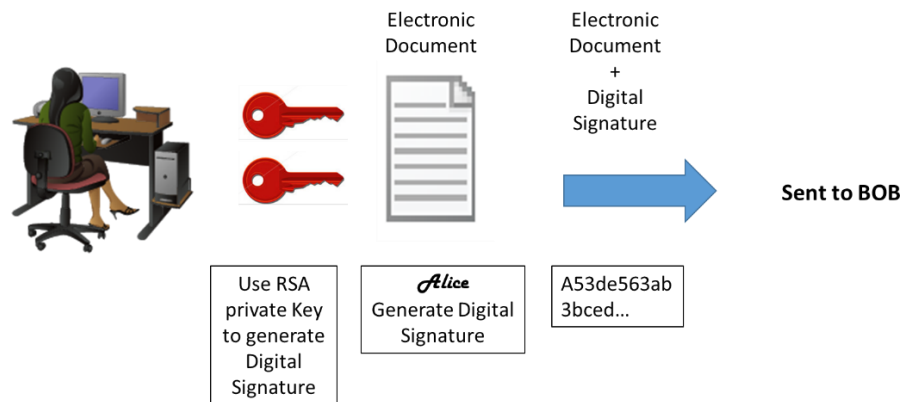
Table 1 - RSA Public and Private Keys

Public RSA Key	d94d889e88853dd89769a18015a0a2e6bf82bf356fe14f251fb4f5e2df0d9f9a94a68a30c428b39e3362fb3779a497ecea37100f264d7fb9fb1a97fbf621133de55fdcb9b1ad0d7a31b379216d79252f5c527b9bc63d83d4ecf4d1d45cbf843e8474babc655e9bb 6799cba77a47eafa838296474afc24beb9c825b73ebf549
Private RSA Key	47b9cfde843176b88741d68cf096952e950813151058ce46f2b048791a26e507a1095793c12bae1e09d82213ad9326928cf7c2350acb19c98f19d32d577d666cd7bb8b2b5ba629d25ccf72a5ceb8a8da038906c84dcd1fe677dff2c029fd8926318ede1b5827 2af22bda5c5232be066839398e42f5352df58848adad11a1

Public Exponent	10001
------------------------	-------

Step 1: Sign the Document.

Alice signs a legal document and send it to Bob using the RSA public and private keys shown in the table above. Now Bob will have to verify Alice's digital signature in order to trust the authenticity of the electronic document.



Step 2: Verify Digital Signature.

Bob receives the document with a digital signature shown in the table below.

Table 2 - Alice's Digital Signature

Alice's Digital Signature
0xc8 0x93 0xa9 0x0d 0x8f 0x4e 0xc5 0xc3 0x64 0xec 0x86 0x9d 0x2b 0x2e 0xc9 0x21 0xe3 0x8b 0xab 0x23 0x4a 0x4f 0x45 0xe8 0x96 0x9b 0x98 0xbe 0x25 0x41 0x15 0x9e 0xab 0x6a 0xfb 0x75 0x9a 0x13 0xb6 0x26 0x04 0xc0 0x60 0x72 0x28 0x1a 0x73 0x45 0x71 0x83 0x42 0xd4 0x7f 0x57 0xd1 0xac 0x91 0x8c 0xae 0x2f 0x3b 0xd2 0x99 0x30 0x3e 0xe8 0xa8 0x3a 0xb3 0x5d 0xfb 0x4a 0xc9 0x18 0x19 0xfd 0x3f0x0c 0x0a 0x1f 0x3d 0xa4 0xa4 0xfe 0x02 0x9d 0x96 0x2f 0x50 0x34 0xd3 0x95 0x55 0xe0 0xb7 0x2a 0x46 0xa4 0x9e 0xae 0x80 0xc9 0x77 0x43 0x16 0xc0 0xab 0xfd 0xdc 0x88 0x95 0x05 0x56 0xdf 0xc4 0xfc 0x13 0xa6 0x48 0xa3 0x3c 0xe2 0x87 0x52 0xc5 0x3f 0x0c 0x0d

Click <http://nmichaels.org/rsa.py> to use the online RSA tool to verify the authenticity of Alice's digital signature.

Table 3 - Online Digital Signature Tool

RSA Encryptor/Decryptor/Key Generator/Cracker

Directions are at the bottom.

Public Modulus
(hexadecimal):

Public Exponent
(hexadecimal):

Private Exponent
(hexadecimal):

Text:

```

d94d889e88853dd89769a18015a0a2e6bf82bf356fe14f251fb4f5e2df0d9f9a94a68a30c428b39e
3362fb3779a497ecea37100f264d7fb9fb1a97fb621133de55fdbcb9b1ad0d7a31b379216d7925
2f5c527b9bc63d83d4ecf4d1d45cbf843e8474babc655e9bb6799cba77a47eafa838296474afc24b
eb9c825b73ebf549
10001
47b9cfde843176b88741d68cf096952e950813151058ce46f2b048791a26e507a1095793c12bae1e
09d82213ad9326928cf7c2350acb19c98f19d32d577d666cd7bb8b2b5ba629d25ccf72a5ceb8a8da
038906c84dcd1fe677dffb2c029fd8926318eede1b58272af22bda5c5232be066839398e42f5352
df58848adad11a1
0xc8 0x93 0xa9 0x0d 0x8f 0x4e 0xc5 0xc3 0x64 0xec 0x86 0x9d 0x2b 0x2e 0xc9 0x21
0xe3 0x8b 0xab 0x23 0x4a 0x4f 0x45 0xe8 0x96 0x9b 0x98 0xbe 0x25 0x41 0x15 0x9e
0xab 0x6a 0xfb 0x75 0x9a 0x13 0xb6 0x26 0x04 0xc0 0x60 0x72 0x28 0x1a 0x73 0x45
0x71 0x83 0x42 0xd4 0x7f 0x57 0xd1 0xac 0x91 0x8c 0xae 0x2f 0x3b 0xd2 0x99 0x30
0x3e 0xe8 0xa8 0x3a 0xb3 0x5d 0xfb 0x4a 0xc9 0x18 0x19 0xfd 0x3f 0x0c 0x0a 0x1f
0x3d 0xa4 0xa4 0xfe 0x02 0x9d 0x96 0x2f 0x50 0x34 0xd3 0x95 0x55 0xe0 0xb7 0x2a
0x46 0xa4 0x9e 0xae 0x80 0xc9 0x77 0x43 0x16 0xc0 0xab 0xfd 0xdc 0x88 0x95 0x05
0x56 0xdf 0xc4 0xfc 0x13 0xa6 0x48 0xa3 0x3c 0xe2 0x87 0x52 0xc5 0x3f 0x0c 0x0d

```

Hexadecimal ☒

Character String ☐

Encrypt

Sign

Decrypt

Verify

Generate

Crack

- Copy and paste the **public** and **private** keys from Table 1 above into the **Public Modulus** and **Private Exponent** boxes on the website as shown in the picture above.
- Make sure the Public Exponent is 10001.
- Paste Alice's digital signature from Table 2 in the box labeled text on the website as shown above.
- Now BOB can verify the digital signature by clicking the **Verify** button near the bottom center of the website. Whose signature is identified?

Alice's name should be displayed.

Step 3 : Generate a Response Signature.

Bob receives and verifies Alice's electronic document and digital signature. Now Bob creates an electronic document and generates his own digital signature using the private RSA Key in Table 1 (Note: Bob's name is in all capital letters).

Table 4 - BOB Digital Signature

BOB's Digital Signature
0x6c 0x99 0xd6 0xa8 0x42 0x53 0xee 0xb5 0x2d 0x7f 0x0b 0x27 0x17 0xf1 0x1b 0x62 0x92 0x7f 0x92 0x6d 0x42 0xbd 0xc6 0xd5 0x3e 0x5c 0xe9 0xb5 0xd2 0x96 0xad 0x22 0x5d 0x18 0x64 0xf3 0x89 0x52 0x08 0x62 0xe2 0xa2 0x91 0x47 0x94 0xe8 0x75 0xce 0x02 0xf8 0xe9 0xf8 0x49 0x72 0x20 0x12 0xe2 0xac 0x99 0x25 0x9a 0x27 0xe0 0x99 0x38 0x54 0x54 0x93 0x06 0x97 0x71 0x69 0xb1 0xb6 0x24 0xed 0x1c 0x89 0x62 0x3d 0xd2 0xdf 0xda 0x7a 0x0b 0xd3 0x36 0x37 0xa3 0xcb 0x32 0xbb 0x1d 0x5e 0x13 0xbc 0xca 0x78 0x3e 0xe6 0xfc 0x5a 0x81 0x66 0x4e 0xa0 0x66 0xce 0xb3 0x1b 0x93 0x32 0x2c 0x91 0x4c 0x58 0xbf 0xff 0xd8 0x97 0x2f 0xa8 0x57 0xd7 0x49 0x93 0xb1 0x62

Bob sends the electronic document and digital signature to Alice.

Step 4: Verify Digital Signature.

- Copy and paste the **public** and **private** keys from Table 1 above into the **Public Modulus** and **PrivateExponent** boxes on the website as shown in the picture above.
- Make sure the Public Exponent is 10001.
- Paste Bob's digital signature from Table 4 in the box labeled text on the website as shown above.
- Now Alice can verify the digital signature by clicking the **Verify** button near the bottom center of the website. Whose signature is identified?

Bob's name should be displayed.

Part 4: Create Your Own Digital Signature

Now that you see how digital signatures work, you can create your own digital signature.

Step 1: Generate a New Pair of RSA Keys.

Go to the website tool and generate a new set of RSA public and private keys

- Delete the contents of the boxes labeled **Public Modulus**, **Private Modulus** and **Text**. Just use your mouse to highlight the text and press the delete key on your keyboard.
- Make sure the "Public Exponent" box has **10001**.

- c) Generate a new set of RSA keys by clicking the **Generate** button near the bottom right of the website.
- d) Copy the new keys in Table 5.

Table 5 - New RSA Keys

Public Key	A string of 256 hexadecimal characters will be displayed for both the public and private keys. The keys will be different from one another.
Private key	A string of 256 hexadecimal characters will be displayed for both the public and private keys. The keys will be different from one another.

- e) Now type in your full name into the box labeled **Text** and click **Sign**.

Table 6 - Personal Digital Signature

Personal Digital Signature	A string of characters with this format will be displayed. 0x23 0x90
-----------------------------------	--

Part 5: Exchange and Verify Digital Signatures

Now you can use this digital signature.

Step 1: Exchange your new public and private keys in Table-5 with your lab partner.

- a) Record your lab partner's public and private RSA keys from their Table-5.
- b) Record both keys in the table below.

Table 7- Lab Partners RSA Keys

Public key	A string of 256 hexadecimal characters will be displayed for both the public and private keys. The keys will be different from one another.
Private key	A string of 256 hexadecimal characters will be displayed for both the public and private keys. The keys will be different from one another.

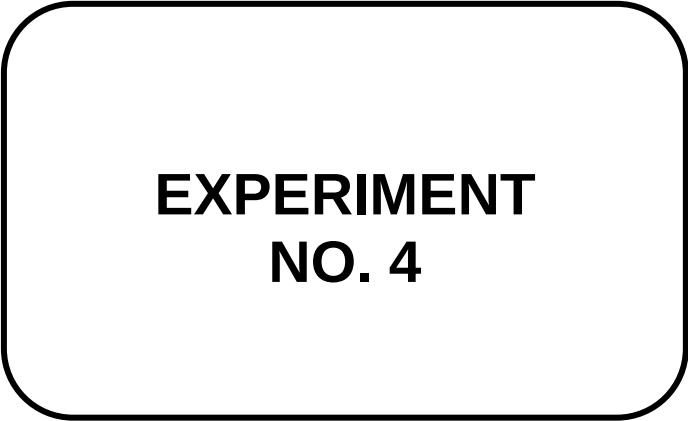
- c) Now exchange their digital signature from their Table-6. Record the digital signature in the table below.

Lab Partner's Digital Signature	A string of characters with this format will be displayed. 0x23 0x90
--	---

Step 2: Verify Lab Partners Digital Signature

- To verify your lab partner's digital signature, paste his or her public and private keys in the appropriate boxes labeled **Public and Private modulus** on the website.
- Now paste the digital signature in the box labeled **Text**.
- Now verify his or her digital signature by clicking the button labeled verify.
- What shows up in the Text box?

Answers will vary.



EXPERIMENT NO. 4

EXPERIMENT NO.4

NAME: -

Backup Data to External Storage.

OBJECTIVE: -

Backup user data.

Part 1: Use a local external disk to backup data.

Part 2: Use a remote disk to backup data.

BACKGROUND/ SCENARIO: -

It is important to establish a backup strategy that includes data recovery of personal files. While many backup tools are available, this lab focuses on the Microsoft Backup Utility to perform backups to local external disks. In Part 2, this lab uses the Dropbox service to backup data to a remote or cloud-based drive.

REQUIRED RESOURCE: -

- PC or mobile device with Internet access

Part 6 : Backing Up to a Local External Disk

Step 1 : Getting Started With Backup Tools in Windows

Computer usage and organizational requirements determine how often data must be backed up and the type of backup to perform. It can take a long time to run a backup. If the backup strategy is followed carefully, it is not necessary to back up all files every time. Only the files that have changed since the last backup need to be backed up.

Microsoft Windows includes backup tools that can be used to backup files. In versions earlier than Windows 8, you could use Backup and Restore to backup your files. Windows 8.1 ships with File History which can be used to back up the files in the Documents, Music, Pictures, Videos, and Desktop folders. Over time, File History builds a history of your files, allowing you to go back and recover specific versions of a file. This is a helpful feature if there are damaged or lost files.

Windows 7 and Vista ship with a different backup tool called **Backup and Restore**. When an external drive is selected, Windows 7 will offer the chance to use the new drive as a backup device. Use Backup and Restore to manage backups.

To access the Backup and Restore utility in Windows 7, follow the steps below:

- a) Connect an external drive.
- b) Execute the Backup and Restore by using the following path:

Start > Control Panel > Backup and Restore

To get started with File History in Windows 8.1, follow the steps below:

- c) Connect an external drive.
- d) Turn on File History by using the following path:

Control Panel > File History > click Turn on

Note: Other operating systems also have backup tools available. Apple OS X includes Time Machine while Ubuntu Linux includes Déjà Dup, by default.

Step 2: Backing up the Documents and Pictures folders

Now that the external disk is connected and you know how to find the backup tool, set it up to back up the Documents and Pictures folders every day, at 3 a.m.

- a) Open **Backup and Restore** (Windows 7) or **File History** (Windows 8.x).
- b) Select the external disk you want to use to receive the backup.
- c) Specify what you want to be backed up to the disk. For this lab, choose the **Documents** and **Pictures** folders.
- d) Set up a backup schedule. For this lab, use daily at 3 a.m. Why would you choose to perform backups at 3 a.m.?

Because this is likely to be a low-usage time and little to no impact to user operations is expected.

- e) Start the backup by clicking the **Save settings and run backup**.

Part 7: Backing Up to a Remote Disk

Step 1: Getting Familiar with Cloud-Based Backup Services

Another option for a backup destination is a remote disk. This might be a complete cloud service, or simply a NAS connected to the network, remote backups are also very common.

- a) List a few of cloud-based backup services.

Answers will vary. Dropbox, Microsoft OneDrive, Google Drive, Apple iCloud, Amazon AWS.

- b) Research the services you listed above. Are these services free?

Answers will vary. Dropbox, Microsoft OneDrive, Google Drive and Apple iCloud are free for a small and limited amount of storage. If the user needs more storage space, a

c) Are the services listed by you platform dependent?

Answers will vary. Most of the popular cloud backup services are not platform dependent and have web interfaces or clients available for all the

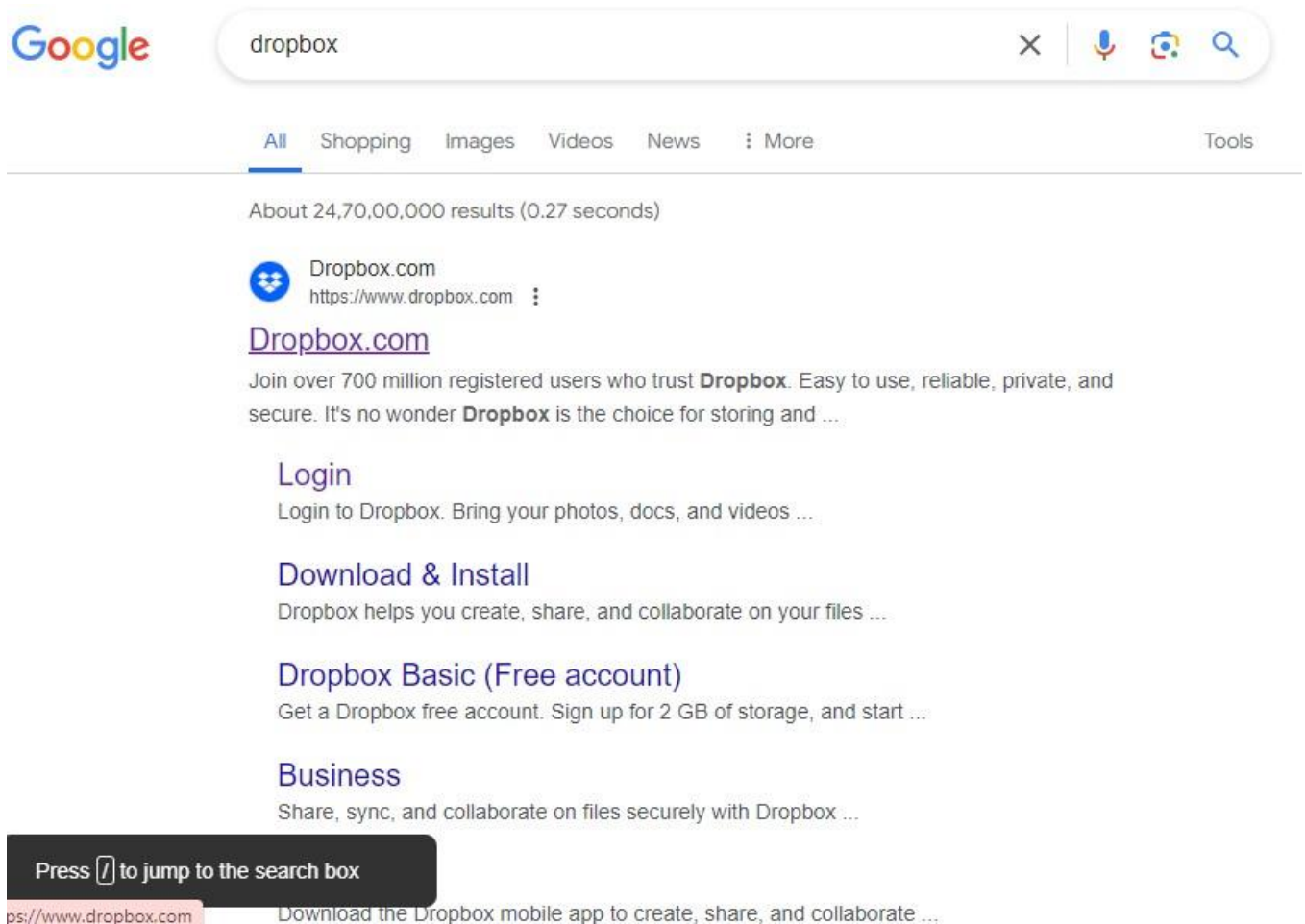
d) Can you access your data from all devices you own (desktop, laptop, tablet and phone)?

Answers will vary. Yes. Most of the popular cloud-based backup services have clients for all computerplatforms

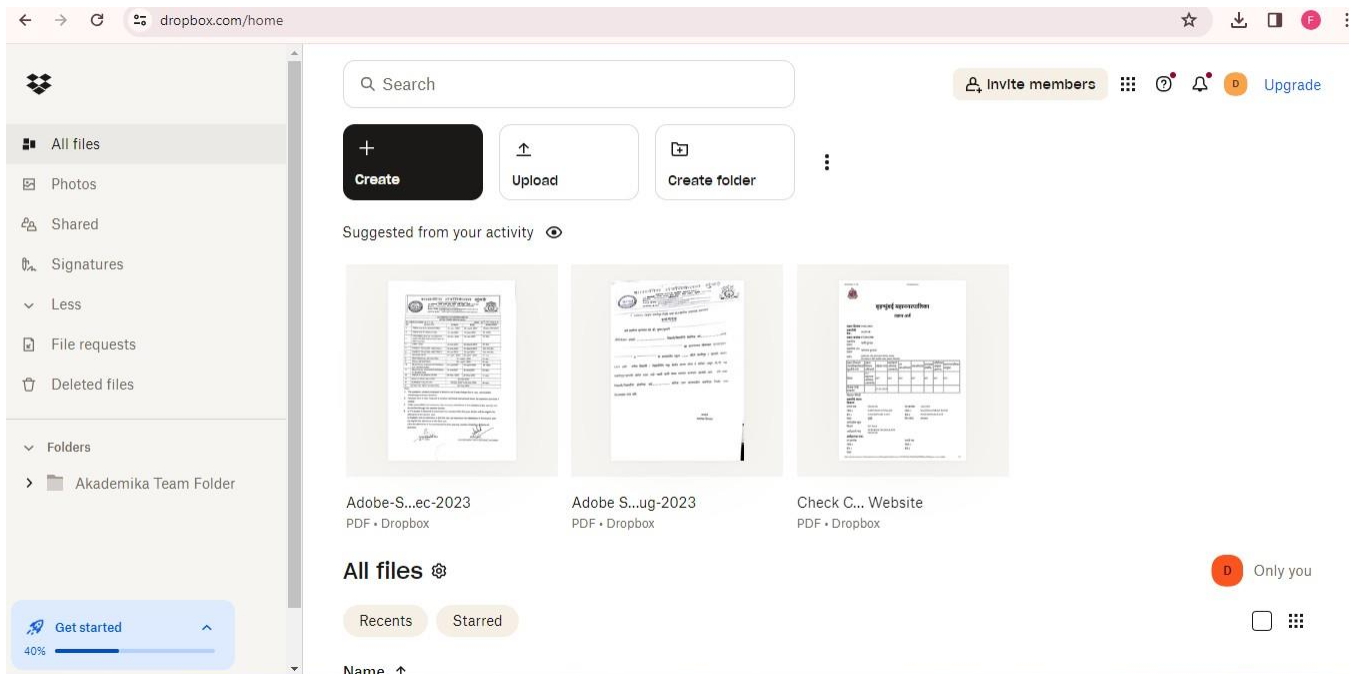
Step 2: Using Backup and Restore to Back Up Data to the Cloud

Choose a service that fits your needs and backup your copy of your Documents folder to the cloud. Notice that Dropbox and OneDrive allow you to create a folder on your computer that acts as a link to the cloud drive. Once created, files copied to that folder are automatically uploaded to the cloud by the cloud-service client that is always running. This setup is very convenient because you can use any backup tools of your choice to schedule cloud backups. To use Windows Backup and Restore to back up your files to Dropbox, follow the steps below:

- a) Visit <http://dropbox.com> and sign up for a free Dropbox account.
- b) Open Google Chrome then search Dropbox.



- c) Click on start a free Trial.
- d) Sign up using your Google account for free.
- e) Click on your Account and then Select MyDropbox.
- f) Select a image that you want to upload on Dropbox cloud service.



- g) When the account is created, Dropbox will display all the files stored in your account. Click **your name** and click **Install** to download and install the appropriate Dropbox client for your operating system.
- h) Open the downloaded program to install the client.
- i) After the installation is complete, the Dropbox client will create a folder named Dropbox inside your Home folder. Notice that any files copied into the newly created folder will be automatically copied to Dropbox's cloud-hosted servers.
- j) Open **Windows Backup and Restore** and configure it to use the new Dropbox folder as a backup destination.

REFLECTION: -

1. What are the benefits of backing up data to a local external disk?

Local backups are entirely under user's control and no other parties have access to the data. Local backups are always available as it does not depend on an Internet connection.

2. What are the drawbacks of backing up data to a local external disk?

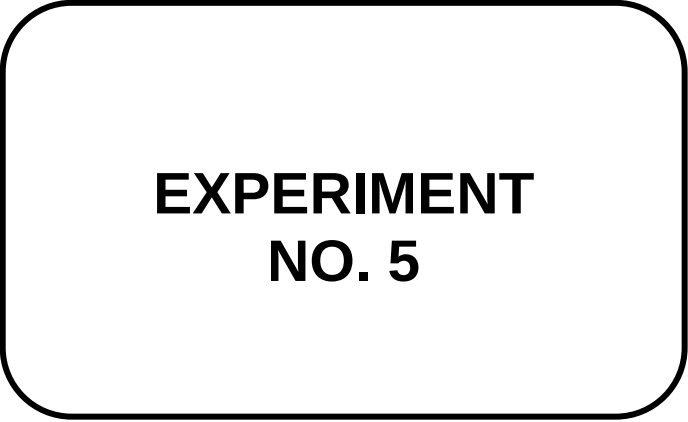
Measures must be taken if the user wants/needs to access the data from other devices or locations. In case of a disaster such as fire or flooding, the local backup disks are also at risk.

3. What are the benefits of backing up data to a cloud-based disk?

Because data is replicated to an off-site location, it is more resilient to disaster.

4. What are the drawbacks of backing up data to a cloud-based disk?

The data is now under the care of a third party company. This company can read and access the data. Also, if the amount of data is high, a storage fee must be paid.



EXPERIMENT NO. 5

EXPERIMENT NO.5

NAME: -

Compare Data with a Hash

OBJECTIVES: -

Use a hashing program to verify the integrity of data.

BACKGROUND/ SCENARIO: -

It is important to understand when data has been corrupted or it has been tampered with. A hashing program can be used to verify if data has changed, or if it has remained the same. A hashing program performs a hash function on data or a file, which returns a (usually much shorter) value. There are many different hash functions, some very simple and some very complex. When the same hash is performed on the same data, the value that is returned is always the same. If any change is performed on the data, the hash value returned will be different.

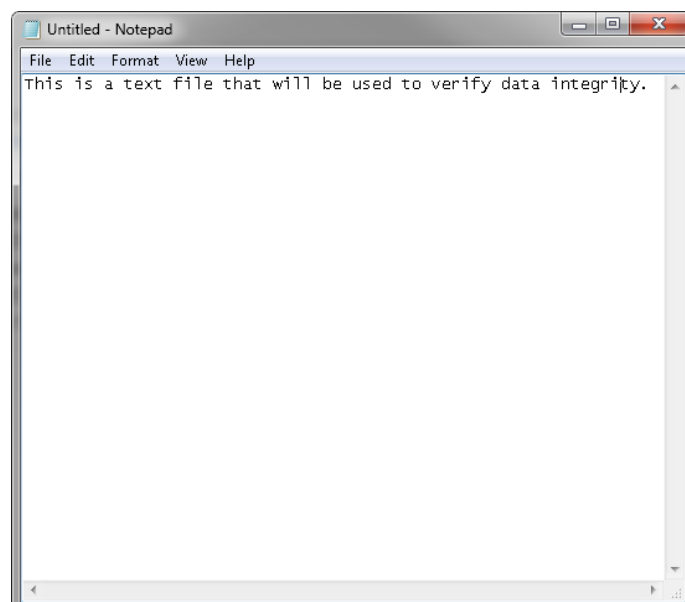
Note: You will need installation privileges and some knowledge of the process to install Windows programs.

REQUIRED RESOURCE: -

- PC with Internet access

Create a Text file

Search your computer for the

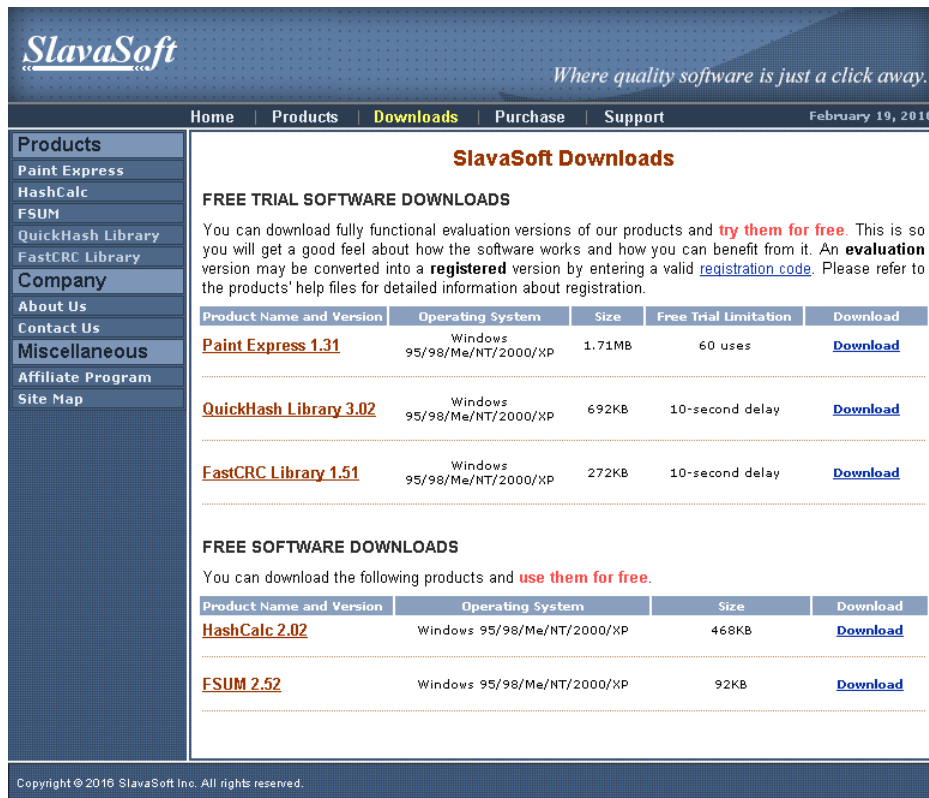


and open it. Type some text in the program.

Choose **File > Save**. Navigate to **Desktop**. Type **Hash** in the **File name:** field, and click **Save**.

Install HashCalc

Open a web browser and navigate to <http://www.slavasoft.com/download.htm>.



SlavaSoft Downloads

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You can download fully functional evaluation versions of our products and **try them for free**. This is so you will get a good feel about how the software works and how you can benefit from it. An **evaluation** version may be converted into a **registered** version by entering a valid [registration code](#). Please refer to the products' help files for detailed information about registration.

Product Name and Version	Operating System	Size	Free Trial Limitation	Download
Paint Express 1.31	Windows 95/98/Me/NT/2000/XP	1.71MB	60 uses	Download
QuickHash Library 3.02	Windows 95/98/Me/NT/2000/XP	692KB	10-second delay	Download
FastCRC Library 1.51	Windows 95/98/Me/NT/2000/XP	272KB	10-second delay	Download

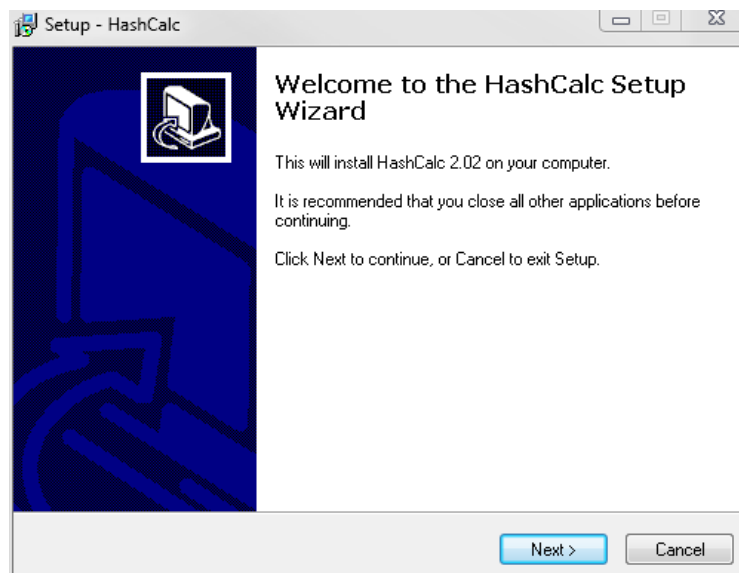
FREE SOFTWARE DOWNLOADS

You can download the following products and **use them for free**.

Product Name and Version	Operating System	Size	Download
HashCalc 2.02	Windows 95/98/Me/NT/2000/XP	468KB	Download
FSUM 2.52	Windows 95/98/Me/NT/2000/XP	92KB	Download

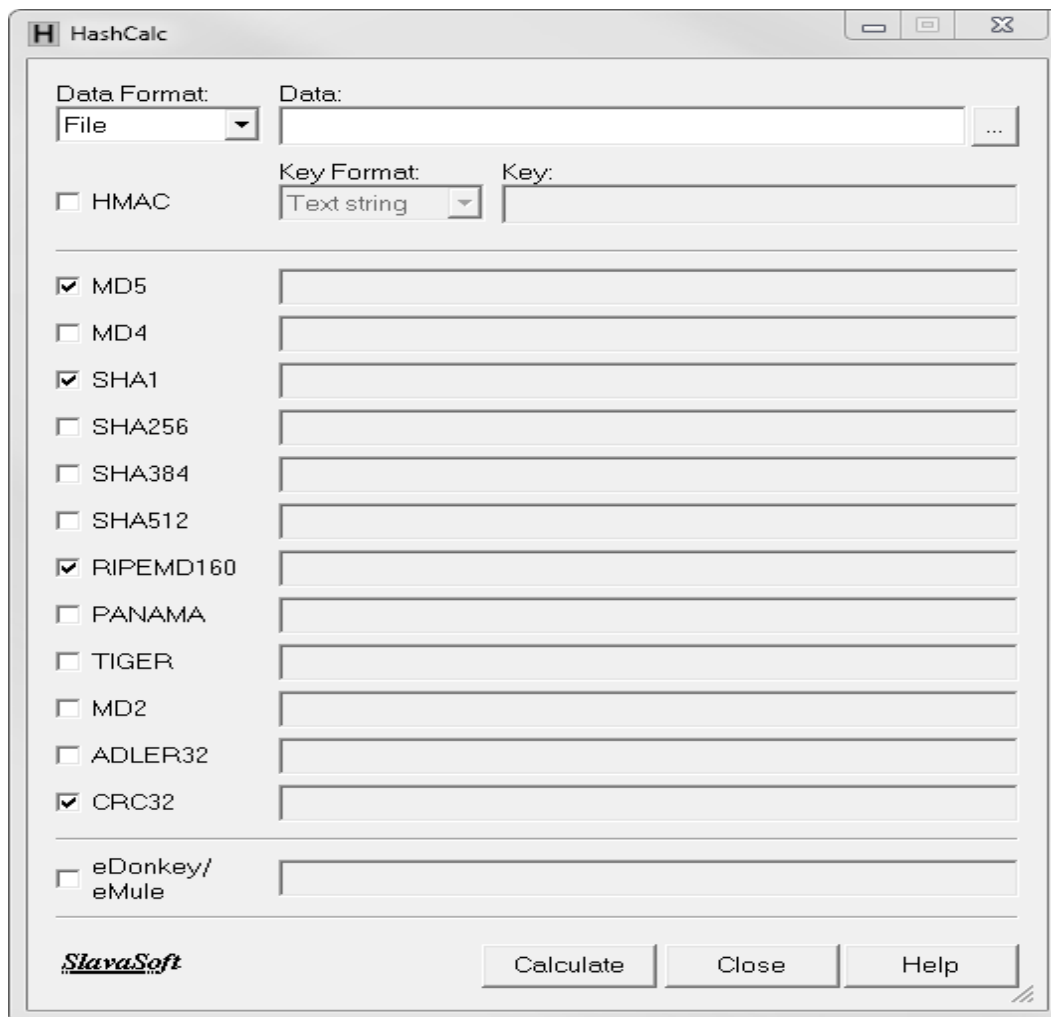
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Click **Download** in the **HashCalc 2.02** row. Open the **hashcalc.zip** file and run the **setup.exe** file inside.



Follow the installation wizard to install HashCalc.

Click **Finish** on the last screen, and close the **README** file if it opened. You may read the file if you wish. HashCalc is now installed and running.



Calculate a hash of the Hash.txt file

Set the following items in HashCalc: Data Format: **File**.

Data: Click the ... button next to the Data field, navigate to the **Desktop** and choose the **Hash.txt** file. Uncheck **HMAC**.

Uncheck all hash types except **MD5**.

Click the **Calculate** button. What is the value next to **MD5**?

Answers may vary. Example: 82e8ac8d4ae929c79e1ce4cf1f0691f4

Make a change to the Hash.txt file

Navigate to the **Desktop** and open the **Hash.txt** file.

Make a minor change to the text, such as deleting a letter, or adding a space or period

Click **File > Save**, and close **Notepad**.

Calculate a new hash of the Hash.txt file

Click the **Calculate** button in HashCalc again. What is the value next to **MD5**?

Answers may vary, but will be different from Step 3. Example:

fb53ad826cefac1dbee8f583a66b7bf4 Is the value different from the value recorded in Step 3?

Yes.

Place a check mark next to all of the hash types. Click **Calculate**.

Notice that many of the hash types create a hash of a different length. Why?

Many of the hashes use a different number of bits to produce the hash.

**EXPERIMENT
NO. 6**

EXPERIMENT NO.6

NAME: -

Install a Virtual Machine on a Personal Computer

OBJECTIVE: -

Part 1: Prepare a Computer for Virtualization

Part 2: Import a Virtual Machine into VirtualBox Inventory

BACKGROUND / SCENARIO: -

Computing power and resources have increased tremendously over the last 10 years. A benefit of having multicore processors and large amounts of RAM is the ability to use virtualization. With virtualization, one or more virtual computers can operate inside a single physical computer. Virtual computers that run within physical computers are called virtual machines. Virtual machines are often called guests, and physical computers are often called hosts. Anyone with a modern computer and operating system can run virtual machines.

A virtual machine image file has been created for you to install on your computer. In this lab, you will download and import this image file using a desktop virtualization application, such as VirtualBox.

REQUIRED RESOURCES: -

- Computer with a minimum of 2 GB of RAM and 8 GB of free disk space
- High speed Internet access to download Oracle VirtualBox and the virtual machine image file

Note: The image file is about 2.5 GB, and can grow up to 5 GB after the virtual machine is in operation. While you can delete the image file after the virtual machine is imported, the 8 GB free disk space requirement is for users who decide to keep the image file.

Note: To install and run 64bit virtual machines on a host physical computer, the computer needs to be a 64bit system and have hardware virtualization technology enabled in BIOS. If you are unable to install the virtual machine image you may need to reboot your computer and enter setup mode in BIOS to enable hardware virtualization technology under advanced system settings.

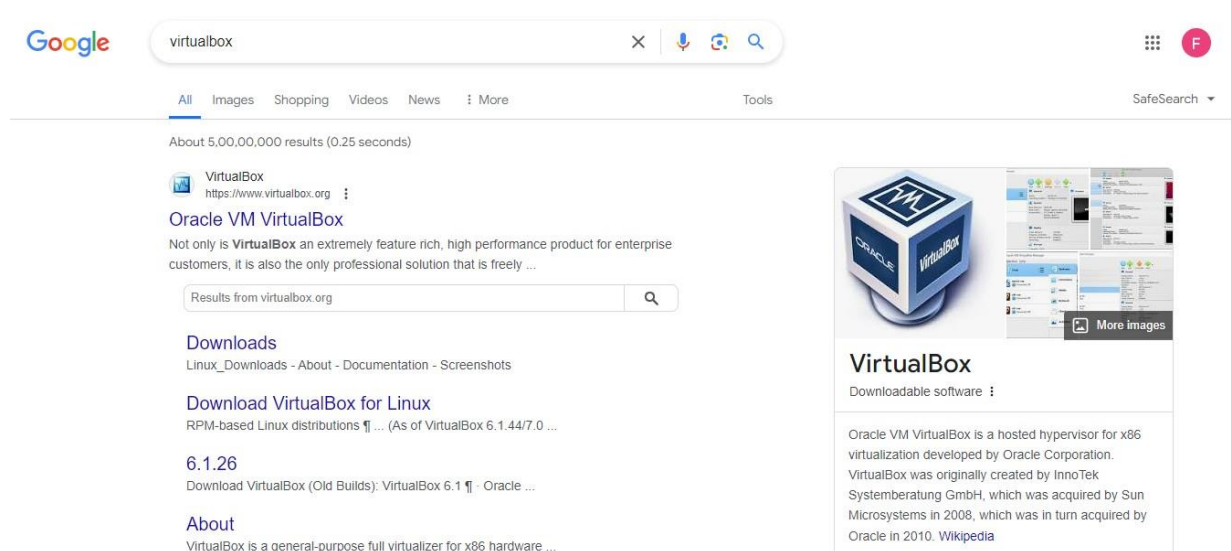
Part 1: Prepare a Host Computer for Virtualization

In Part 1, you will download and install desktop virtualization software, and also download an image file that can be used to complete labs throughout the course. For this lab, the virtual machine is running Linux.

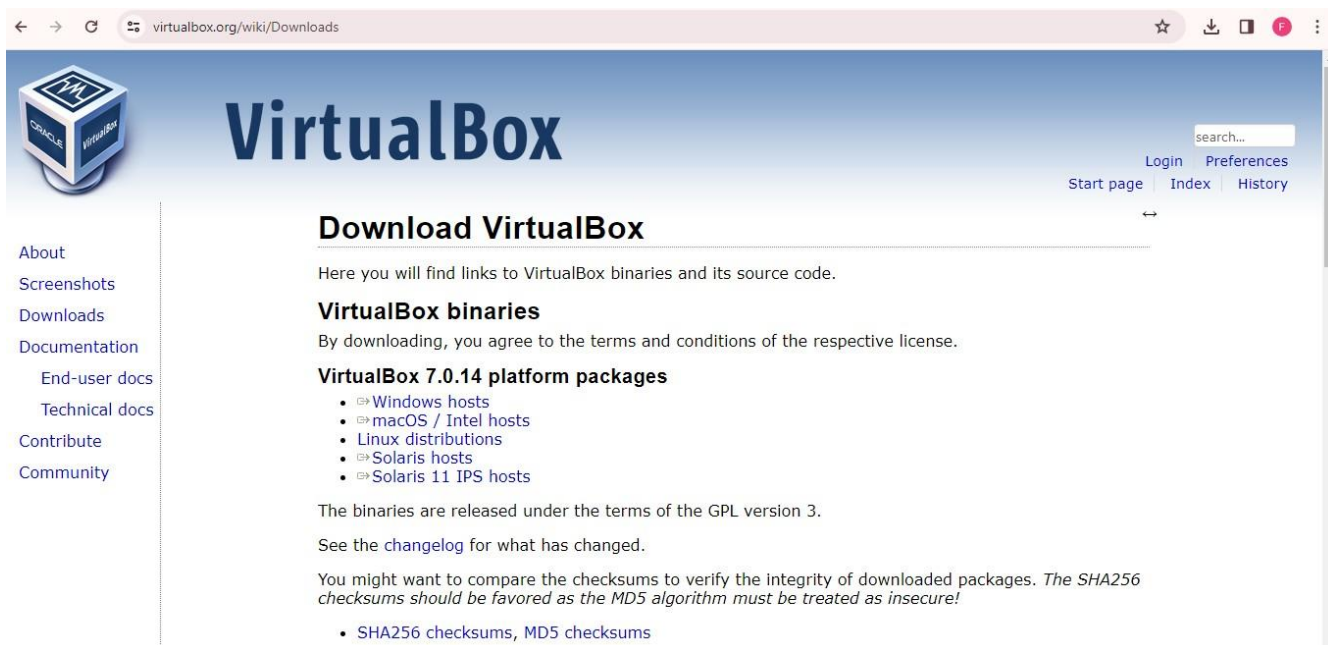
Step 1: Download and install VirtualBox

VMware Workstation Player and Oracle VirtualBox are two virtualization programs that you can download and install to support the image file. In this lab, you will use VirtualBox.

- a) Navigate to http://www.oracle.com/technetwork/server_storage/virtualbox/downloads/index.html.



- b) Choose and download the appropriate installation file based on your operating system.



- c) If your operating system is windows then select Windows Hosts. In case your Operating System kali Linux or Ubuntu you should select Linux Distributions.
- d) When you have downloaded the VirtualBox installation file, run the installer and accept the default installation settings.
- e) After Downloading Virtual Box Application, you will get message on your screen showing that 'Do you want the following program make changes in your computer?'

- f) Then after, for installing VirtualBox on your computer click on Yes.
- g) After clicking on yes you will be entered in next steps of installation of Virtualbox .



- h) Click on Next.
- i) For installing Virtualbox you might get disconnected your Network Connection Temporarily and reset your Network Connection Features. Click on Yes.
- j) Wait for a while to complete your Virtualbox Setup.

Step 2: Download the Virtual Machine image file

The image file was created in accordance with the Open Virtualization Format (OVF). OVF is an open standard for packaging and distributing virtual appliances. An OVF package has several files placed into one directory. This directory is then distributed as an OVA package. This package contains all of the OVF files necessary for the deployment of the virtual machine. The virtual machine used in this lab was exported in accordance with the OVF standard.

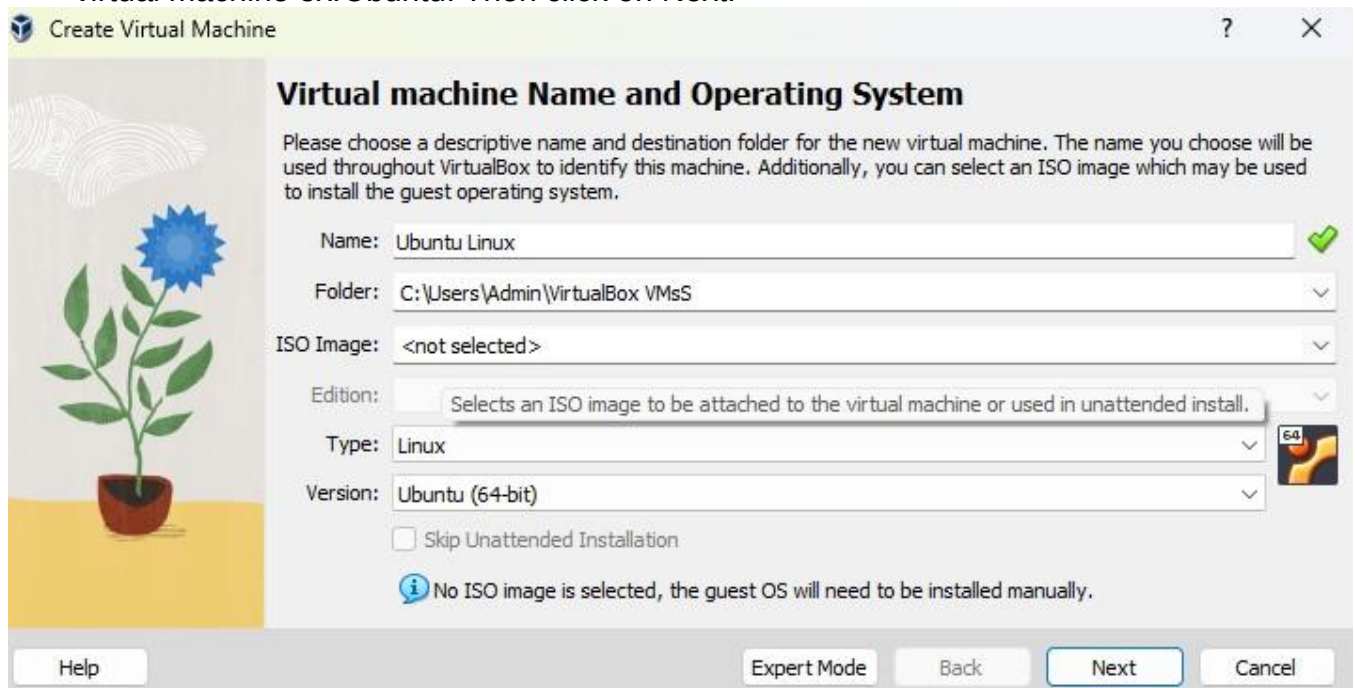
Note: This file is 2.5 GB in size, and it may take over an hour to download, depending on the speed of your Internet connection.

Part 2: Import the Virtual Machine into the VirtualBox Inventory

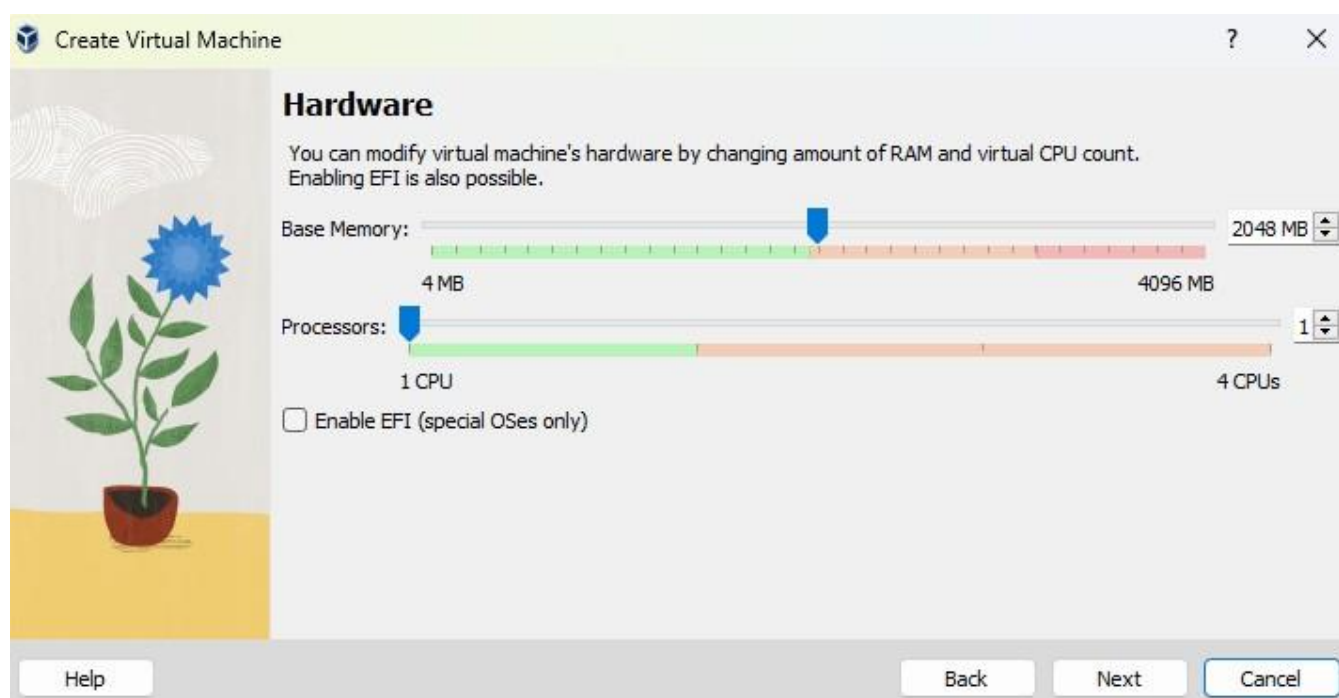
In Part 2, you will import the virtual machine image into VirtualBox and start the virtual machine.

Step 1: Import the virtual machine file into Virtual Box

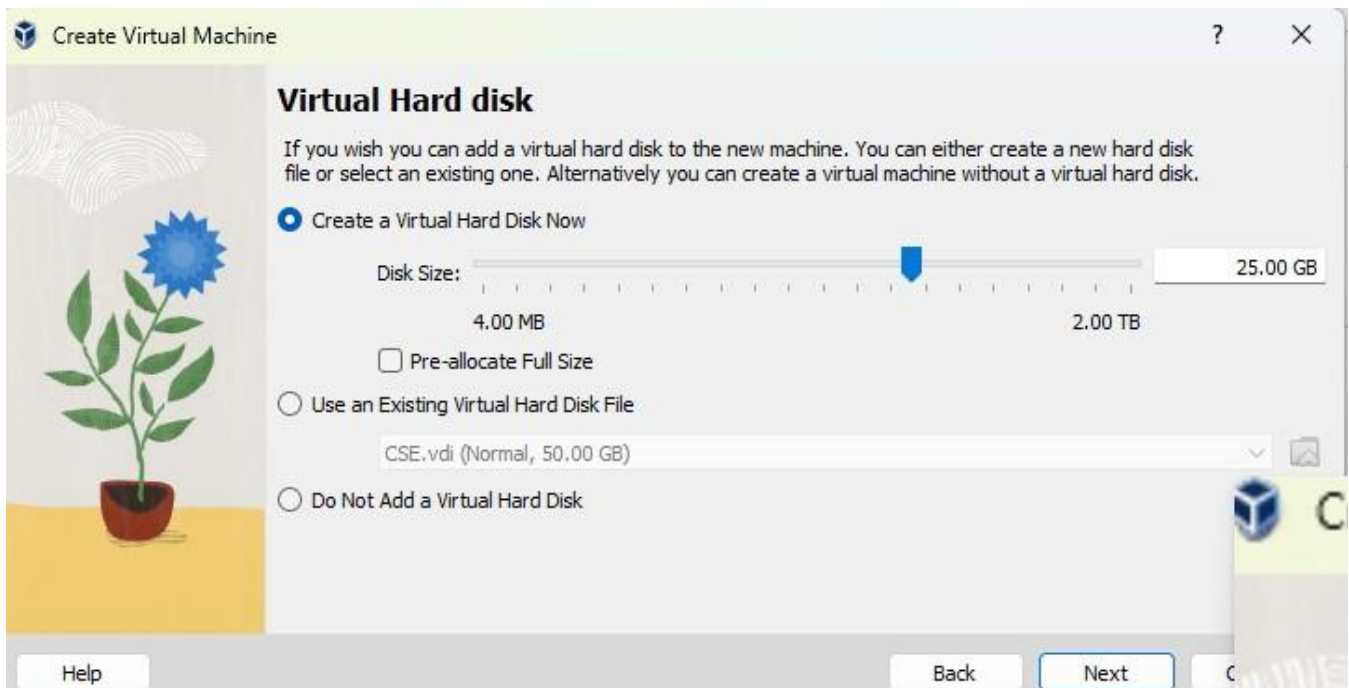
- a) Open **VirtualBox**. Click on New. Specify the File Name that you want to give to your virtual machine ex. Ubuntu. Then click on Next.



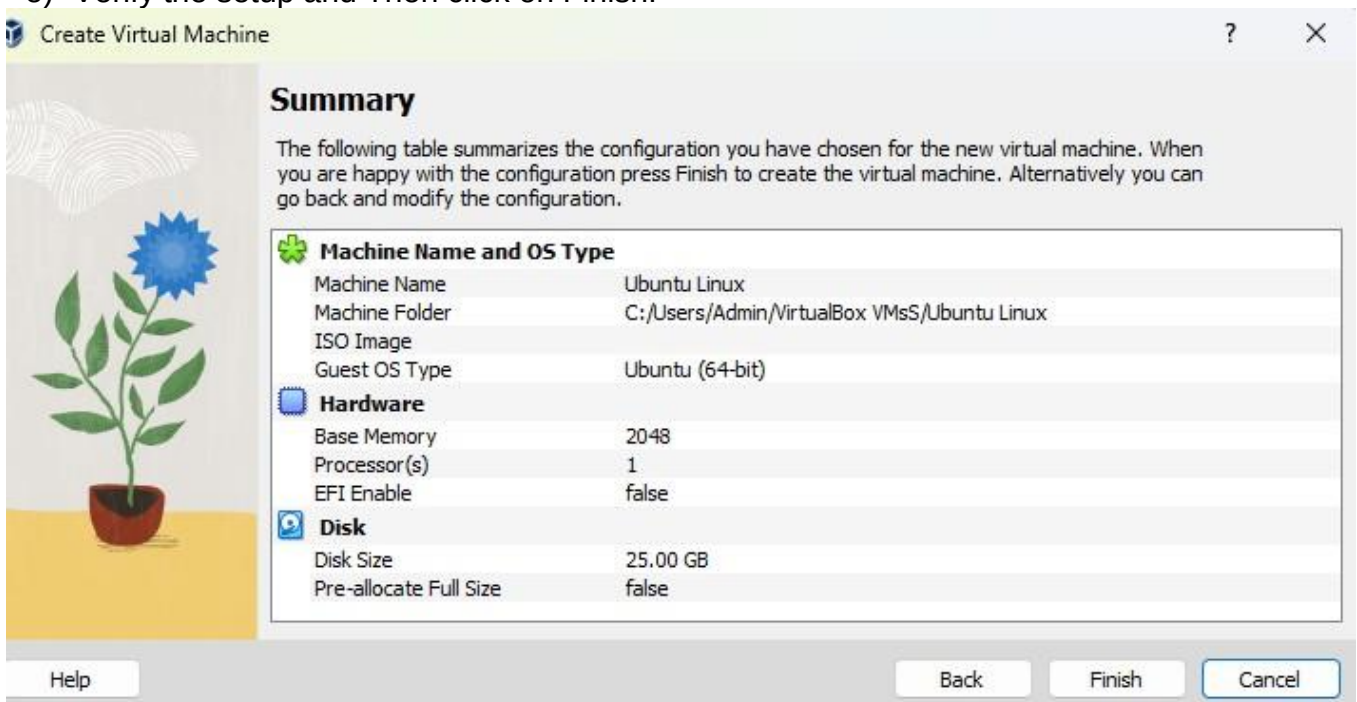
- b) Now you will be asked to create a virtual machine's hardware by allocating the amount of RAM AND Processors according to your requirement. Then click on Next.



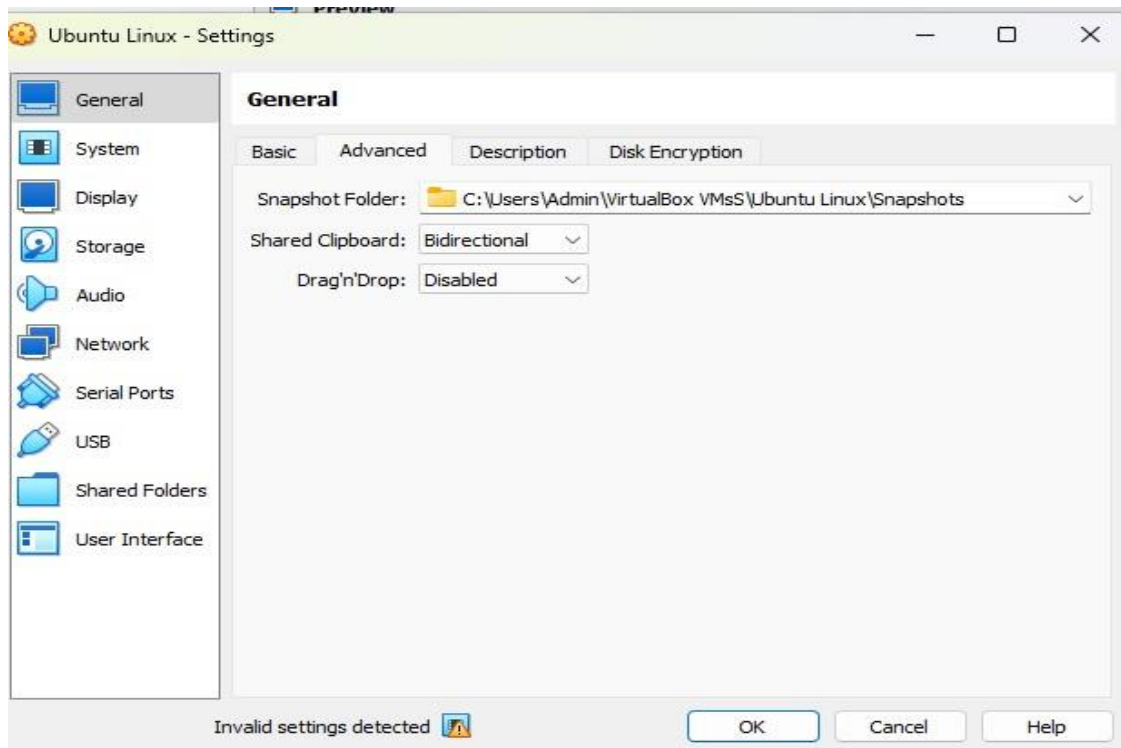
- c) Now set the Username, Password, Host name, Domain Name that you want to give to your virtualMachine. Click on Next.
- d) Create a new hard disk file or select an existing one. Click on create a virtual hard disk Now. And allocate the space according to the requirement of your operating system



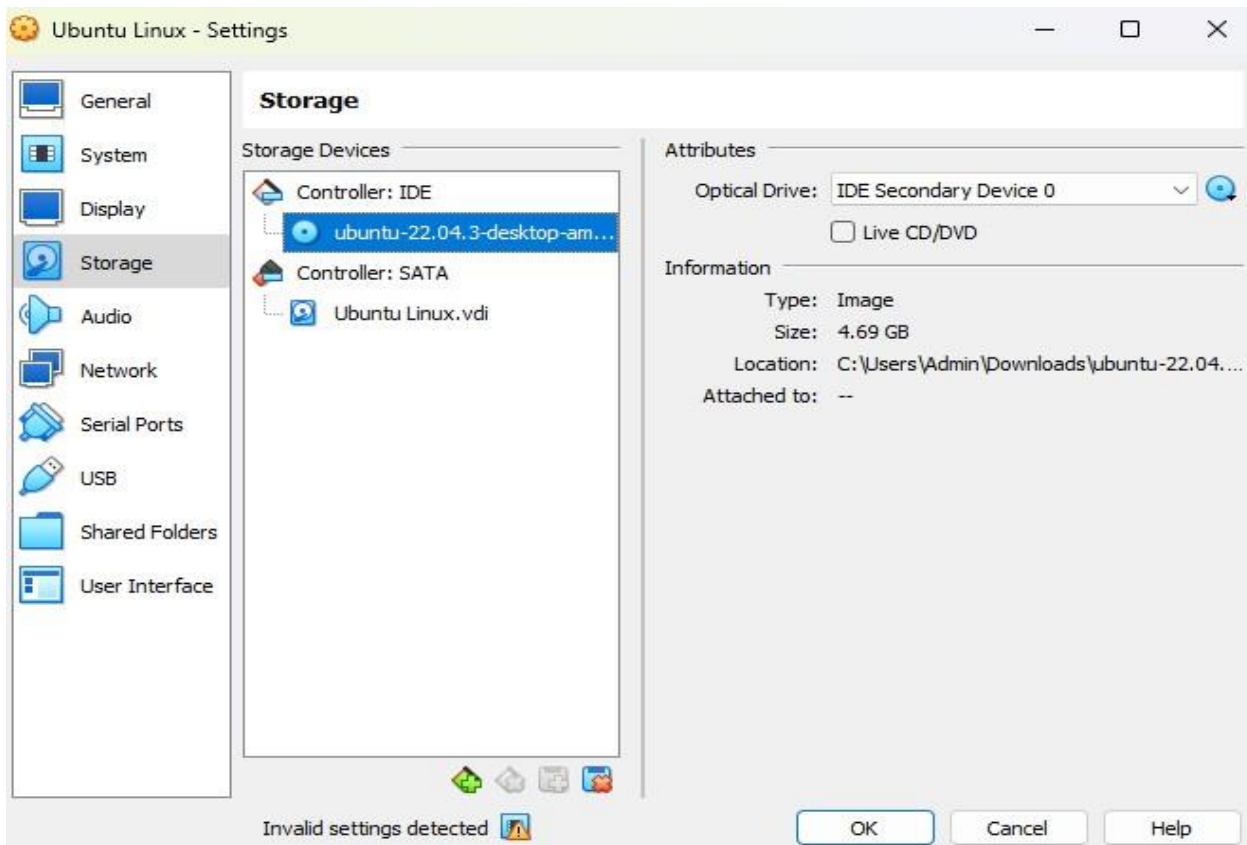
- e) Verify the setup and Then click on Finish.



- f) Now you will be able to see the file that you have created.
- g) Now Click on Settings. In General option click on Advanced, in the snapshot folder select the path for the ISO image file. In the shared clipboard option select the option bidirectional. And in the Drag and Drop select
- h) the Disabled option. Then click on OK



- i) Now click on the Storage and select the ISO image file in the controller IDE option select the ISO image file that you have downloaded and the attributes partition select ISO image file and then click on OK



Step 2: Familiarize yourself with the Virtual Machine.

Use the Ubuntu CyberEss virtual machine you just installed to complete the labs that require Ubuntu in this course. Familiarize yourself with the icons in the list below:

The launcher icons are on the left (from top to bottom):

- Search Tool
- File manager application
- Firefox Web Browser
- LibreOffice Writer, LibreOffice Calc, LibreOffice Impress
- Ubuntu Software Center
- Amazon
- System Settings
- Terminal
- Trash

- a) Open the terminal application. Type the command **ip address** at the prompt to determine the IP address of your virtual machine.

What are the IP addresses assigned to your virtual machine?

Answer will vary. The loopback interface is assigned 127.0.0.1/8, and the Ethernet interface is assigned an IP address in the 10.0.2.15/24 network.

- b) Locate and launch the web browser application. Can you navigate to your favorite search engine?
-

Yes

- c) Press the right ctrl key to release the cursor from the virtual machine. Now go to the menu at the top of the virtual machine window and choose **File > Close** to close the virtual machine. What options are available?
-

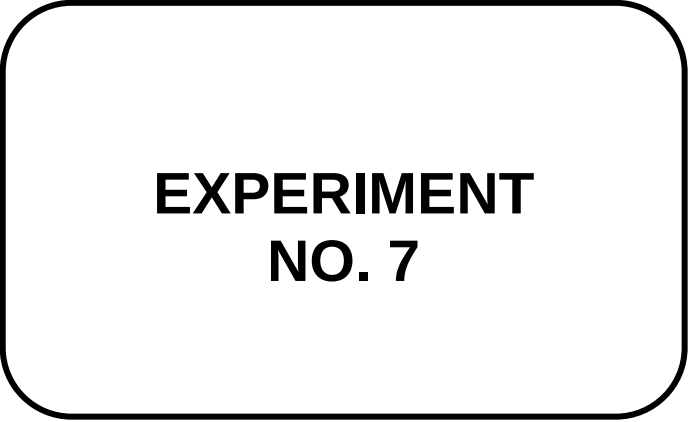
Save the machine state, Send the shutdown signal and power off the machine

- d) Click the **Save the machine state** radio button and click **OK**. The next time you start the virtual machine, you will be able to resume working in the operating system in its current state.

Reflection: -

What are the advantages and disadvantages of using a virtual machine?

With a virtual machine, you are able to test new applications or operating systems without affecting your host machine. You are also able to save the current machine state when you close virtual machine. If you have any issues, you have the option to revert the virtual machine to a previously saved state. machine requires hardware resources from the host machine, such as hard drive space, RAM, and processing power.



EXPERIMENT NO. 7

EXPERIMENT NO.7

NAME: -

Authentication, Authorization, and Accounting.

OBJECTIVE: -

- Given a scenario, select the appropriate authentication, authorization, or access control.
- Install and configure security controls when performing account management, based on best practices.

Part 1: Adding Groups, Users, and Passwords on a Linux System

Part 2: Verify Users, Groups, and Passwords

Part 3: Using Symbolic Permissions **Part 4: Absolute Permissions**

BACKGROUND/ SCENARIO: -

You will be conducting host security practices using the Linux command line by performing the following tasks:

- Adding Groups, Users, and Passwords
- Verifying Groups, Users, and Passwords
- Setting Symbolic Permissions
- Setting Absolute Permissions

REQUIRED RESOURCES: -

- PC with Ubuntu 22.04. LTS installed in a VirtualBox or VMware virtual machine.

Part 1 : Adding Groups, Users, and Passwords on a Linux System

In this part, you will add users, groups, and passwords to the local host machine.



Click on the **terminal** icon to open a terminal. Step 2: Escalate privileges to the root level by entering the `sudo su` command. Enter the password when prompted.

cisco@ubuntu:~\$ `sudo su`

```
cisco@ubuntu:~$ sudo su
[sudo] password for cisco:
root@ubuntu:/home/cisco#
```

Step 3 : Add a new group named HR by entering the command `groupadd HR`.

root@ubuntu:/home/cisco# `groupadd HR`

```
root@ubuntu:/home/cisco# groupadd HR
root@ubuntu:/home/cisco#
```

Part 2: Verify Users, Groups, and Passwords

Step 1: Verify the new group has been added to the group file list by entering `cat /etc/group`.

root@ubuntu:/home/cisco# `cat /etc/group`

```
root@ubuntu:/home/cisco# cat /etc/group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:syslog,cisco
Bob:x:1002:
Eve:x:1003:
Eric:x:1004:
HR:x:1005:
root@ubuntu:/home/cisco#
```

The new group HR will be added to the bottom of the `/etc/group` file with a group ID of 1005.

Step 2: Add a new user named jenny.

root@ubuntu:/home/cisco# `adduser jenny`

- When prompted for a new password, type **lasocial**. Press **Enter**.
- When prompted again, type **lasocial**. Press **Enter**.
- When prompted for a full name, type **Jenny**. Press **Enter**.
- For the rest of the configurations, press **Enter** until when asked is the information correct.
- Type **Y** for yes and press **Enter**.

```
root@ubuntu:/home/cisco# adduser jenny
Adding user `jenny' ...
Adding new group `jenny' (1006) ...
Adding new user `jenny' (1005) with group `jenny' ...
Creating home directory `/home/jenny' ...
Copying files from `/etc/skel' ...
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
Changing the user information for jenny
Enter the new value, or press ENTER for the default
    Full Name []: Jenny
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] Y
```

Step 3: Place the user jenny in the HR group.

```
root@ubuntu:/home/cisco# usermod -G HR jenny
```

```
root@ubuntu:/home/cisco# usermod -G HR jenny
root@ubuntu:/home/cisco#
```

Step 4: Add another new user named joe.

```
root@ubuntu:/home/cisco# adduser joe
```

- When prompted for a new password, type **tooth**. Press **Enter**.
- When prompted again, type **tooth**. Press **Enter**.
- When prompted for a full name, type **Joe**. Press **Enter**.
- For the rest of the configurations, press **Enter** until when asked is the information correct.
- Type **Y** for yes and press **Enter**.

```
root@ubuntu:/home/cisco# adduser joe
Adding user `joe' ...
Adding new group `joe' (1007) ...
Adding new user `joe' (1006) with group `joe' ...
Creating home directory `/home/joe' ...
Copying files from `/etc/skel' ...
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
Changing the user information for joe
Enter the new value, or press ENTER for the default
    Full Name []: Joe
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] Y
```

- Place the user joe in the HR group.

```
root@ubuntu:/home/cisco# usermod -G HR joe
```

```
root@ubuntu:/home/cisco# usermod -G HR joe
root@ubuntu:/home/cisco#
```

Step 5: Verify the newly created users in the passwd file.

```
root@ubuntu:/home/cisco# cat /etc/passwd
```

```
root@ubuntu:/home/cisco# cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
eve:x:1005:1005::/home/eve:/bin/bash
eric:x:1004:1004::/home/eric:/bin/bash
jenny:x:1005:1006:Jenny,,,:/home/jenny:/bin/bash
joe:x:1006:1007:Joe,,,:/home/joe:/bin/bash
```

Step 6: View the created users in the shadow file.

```
root@ubuntu:/home/cisco# cat /etc/shadow
```

Part 3: Using Symbolic Permissions

Step 1: While on the Ubuntu system, press and hold the keys CTRL+ALT+F1 until the screen changes to the tty1 Terminal.

```
Ubuntu 16.04 LTS ubuntu tty1
ubuntu login:
```

Note: If you are unable to use tty1 terminal, return to graphical user interface (GUI) of the host by using **CTRL+ALT+F7** and open a terminal window in the GUI Ubuntu OS. At the prompt, enter **su -l jenny** at the prompt and enter the password **lasocial**. Proceed to Step 4.

```
cisco@ubuntu:~$ su -l jenny
```

```
cisco@ubuntu:~$ su -l jenny
Password:
jenny@ubuntu:~$
```

Note: If CTRL+ALT+F7 did not work, try CTRL+ALT+F8.

Step 2: Once on the Terminal login screen, type jenny and press Enter.

Step 3: When prompted for the password, type lasocial and press Enter.

Step 4: After a successful login, you will see the *jenny@ubuntu:~\$* prompt.

```
Ubuntu 16.04 LTS ubuntu tty1
ubuntu login: jenny
Password:
Welcome to Ubuntu 16.04 LTS (GNU/Linux 4.4.0-24-generic x86_64)

 * Documentation:  https://help.ubuntu.com/

15 packages can be updated.
0 updates are security updates.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

jenny@ubuntu:~$
```

Since we are not logged in as the *root* (superuser), we are presented with the dollar sign instead of the # if we were to be logged in as the user *root*.

Step 5: View your present directory.

jenny@ubuntu:~\$ pwd

```
jenny@ubuntu:~$ pwd
/home/jenny
```

Step 6: Go back one directory level to the */home* directory.

jenny@ubuntu:~\$ cd ..

```
jenny@ubuntu:~$ cd ..
jenny@ubuntu:/home$
```

Step 7: List all directories and their permissions.

jenny@ubuntu:/home\$ ls -l

```
jenny@ubuntu:/home$ ls -l
total 12
drwxr-xr-x 17 cisco cisco 4096 Jun 28 18:04 cisco
drwxr-xr-x  3 jenny jenny 4096 Jun 28 23:28 jenny
drwxr-xr-x  2 joe   joe   4096 Jun 28 19:18 joe
jenny@ubuntu:/home$
```

The Linux operating system has a total of 10 letters or dashes in the permissions fields:

- The first field is a dash for a file and a d for a directory
- The 2nd through 4th fields are for the user
- The 5th through 7th fields are for the group
- The 8th through 10th fields are for others (accounts other than those in the group)



Step 8: Enter Joe's folder as Jenny by typing the command `cd joe`.

```
jenny@ubuntu:/home$ cd joe
```

```
jenny@ubuntu:/home$ cd joe
jenny@ubuntu:/home/joe$
```

Notice that we are able to go into *Joe's home folder*.

```
jenny@ubuntu:/home/joe$ cd..
```

```
jenny@ubuntu:/home/joe$ cd ..
jenny@ubuntu:/home$
```

Step 9: Press and hold CTRL+ALT+F2 to switch to another Terminal session (tty2).

```
Ubuntu 16.04 LTS ubuntu tty2
ubuntu login: _
```

Step 10: Login as the user root with the password `secretpassword`.

```
Ubuntu 16.04 LTS ubuntu tty2
ubuntu login: root
Password:
Welcome to Ubuntu 16.04 LTS (GNU/Linux 4.4.0-24-generic x86_64)

* Documentation:  https://help.ubuntu.com/

15 packages can be updated.
0 updates are security updates.
```

Note: If you are unable to use tty2 terminal, return to graphical user interface (GUI) of the host by using **CTRL+ALT+F7** and open a terminal window in the GUI Ubuntu OS. At the prompt, enter **sudo -i** at the prompt and enter the password **password**.

```
cisco@ubuntu:~$ sudo -i
[sudo] password for cisco:
root@ubuntu:~#
```

Step 11: Change to the /home directory.

```
root@ubuntu:~# cd /home
```

```
root@ubuntu:~# cd /home
root@ubuntu:/home#
```

Step 12: Change the “other” permission on joe’s folder by making it non-executable.

```
root@ubuntu:/home# chmod o-x joe
```

```
root@ubuntu:/home# chmod o-x joe
root@ubuntu:/home#
```

Step 13: List the directories once more with their respective permissions.

```
root@ubuntu:/home# ls -l
```

```
root@ubuntu:/home# ls -l
total 12
drwxr-xr-x 17 cisco cisco 4096 Jun 28 18:04 cisco
drwxr-xr-x  3 jenny jenny 4096 Jun 28 23:52 jenny
drwxr-xr--  2 joe  joe  4096 Jun 28 19:18 joe
root@ubuntu:/home#
```

Notice now that there are two dashes in the “others” field for joe’s folder.

Step 14: Press and hold CTRL+ALT+F1 to switch back to the other Terminal session (tty1). Make sure you are viewing the following command prompt : jenny@ubuntu:/home\$.**Step 15: Attempt to go into Joe’s folder once more.**

```
jenny@ubuntu:/home$ cd joe
```

```
jenny@ubuntu:/home$ cd joe
-bash: cd: joe: Permission denied
jenny@ubuntu:/home$
```

Notice that we do not have the permissions to do so.

The chart below shows examples of other ways the **chmod** command can be used:

chmod command	Results
chmod u+rwx	Adds read, write, and execute permissions for the user
chmod u+rw	Adds read and write permission for the user
chmod o+r	Adds read permission for others
chmod g-rwx	Removes read, write, and execute permissions for the group

Step 16: Type exit followed by pressing Enter to logout of the Terminal session.

Part 4: Absolute Permissions

Step 1: Login as the user joe with the password tooth while on tty1.

```
Ubuntu 16.04 LTS ubuntu tty1
ubuntu login: joe
Password:
Welcome to Ubuntu 16.04 LTS (GNU/Linux 4.4.0-24-generic x86_64)

* Documentation:  https://help.ubuntu.com/
```

Note: If you are unable to use tty1 terminal, return to graphical user interface (GUI) of the host by using **CTRL+ALT+F7** and open a terminal window in the GUI Ubuntu OS. At the prompt, enter **sudo -l joe** at the prompt and enter the password **tooth**.

```
jenny@ubuntu:/home$ exit
logout
cisco@ubuntu:~$ su -l joe
Password:
joe@ubuntu:~$
```

Step 2: Print your current working directory.

```
joe@ubuntu:~$ pwd
```

```
joe@ubuntu:~$ pwd
/home/joe
joe@ubuntu:~$
```

Step 3: Go back one directory level to the /home directory.

```
joe@ubuntu:~$ cd ..
```

```
joe@ubuntu:~$ cd ..
joe@ubuntu:/home$
```

Step 4: List all directories and their permissions in the current working directory.

```
joe@ubuntu:/home~$ ls -l
```

```
joe@ubuntu:/home$ ls -l
total 12
drwxr-xr-x 17 cisco cisco 4096 Jun 28 18:04 cisco
drwxr-xr-x  3 jenny jenny 4096 Jun 28 23:52 jenny
drwxr-xr--  3 joe  joe  4096 Jun 29 00:12 joe
joe@ubuntu:/home$
```

Notice that Joe's folder is set so that "others" are not able to access the folder.

The other way of assigning permissions besides using symbolic permissions is the use of absolute permissions. Absolute permissions use a three digit octal number to represent the permissions for owner, group and other.

The table below outlines each absolute value and its corresponding permissions:

Number	Permissions
7	Read, Write, and Execute
6	Read and Write
5	Read and Execute
4	Read
3	Write and Execute
2	Write
1	Execute
0	None

By typing the command **chmod 764 examplefile**, the examplefile will be assigned the follow permissions:

- The user will get read, write and execute permissions
- The group will get read and write permissions
- Others will get read access

Breakdown of how 764 represents these permissions:

Digit	Binary Equivalent	Permission
7 (user)	111	1-Read 1-Write1-Execute
6 (group)	110	1-Read 1-Write0-No Execute
4 (others)	100	1-Read 0-No Write 0-No Execute

Step 5: Modify the “others” field for Joe’s folder so that others will be able read and execute butnot write while still maintaining the “user” field to read, write, and execute.

joe@ubuntu:/home\$ **chmod 705 joe**

```
joe@ubuntu:/home$ chmod 705 joe
joe@ubuntu:/home$
```

Step 6: List the file permissions of the current directory to see that the absolute changes were made.

joe@ubuntu:/home\$ ls -l

```
joe@ubuntu:/home$ ls -l
total 12
drwxr-xr-x 17 cisco cisco 4096 Jun 28 18:04 cisco
drwxr-xr-x  3 jenny jenny 4096 Jun 28 23:52 jenny
drwx--r-x  3 joe  joe  4096 Jun 29 00:12 joe
joe@ubuntu:/home$
```

Step 7: Change to the /home/joe directory.

joe@ubuntu:/home\$ cd joe

```
joe@ubuntu:/home$ cd joe
joe@ubuntu:~$
```

Step 8: Create a simple text file named test.txt using touch.

joe@ubuntu:~\$ touch test.txt

```
joe@ubuntu:~$ touch test.txt
joe@ubuntu:~$
```

- Type **exit** followed by pressing **Enter** to log out of Joe's session.
- While on the tty1 Terminal, log back in as **jenny** and enter the password **lasocial**. Press **Enter**.

```
Ubuntu 16.04 LTS ubuntu tty1
ubuntu login: jenny
Password:
```

Note: If you are unable to use tty1 terminal, return to graphical user interface (GUI) of the host by using **CTRL+ALT+F7** and open a terminal window in the GUI Ubuntu OS. At the prompt, enter **su -l jenny** at the prompt and enter the password **lasocial**.

cisco@ubuntu:~\$ su -l jenny

```
joe@ubuntu:~$ exit
logout
cisco@ubuntu:~$ su -l jenny
Password:
jenny@ubuntu:~$
```

Step 9: Change to the /home directory.

jenny@ubuntu:~\$ cd /home

```
jenny@ubuntu:~$ cd /home
jenny@ubuntu:/home$
```

Step 10: List all directories with their respective permissions.

```
jenny@ubuntu:/home$ ls -l
```

```
jenny@ubuntu:/home$ ls -l
total 12
drwxr-xr-x 17 cisco cisco 4096 Jun 28 18:04 cisco
drwxr-xr-x  3 jenny jenny 4096 Jun 28 23:52 jenny
drwx---r-x  3 joe  joe  4096 Jun 29 00:32 joe
jenny@ubuntu:/home$
```

Step 11: Change to the /home/joe directory and list the content of the directory.

```
jenny@ubuntu:/home$ cd joe
jenny@ubuntu:/home/joe$ ls -l
```

```
jenny@ubuntu:/home$ cd joe
jenny@ubuntu:/home/joe$ ls -l
total 12
-rw-r--r-- 1 joe joe 8980 Jun 28 19:18 examples.desktop
-rw-rw-r-- 1 joe joe   0 Jun 29 00:22 test.txt
jenny@ubuntu:/home/joe$
```

Notice that we are able to enter Joe's folder and read the files within the directory. We are able to see the *test.txt* file.

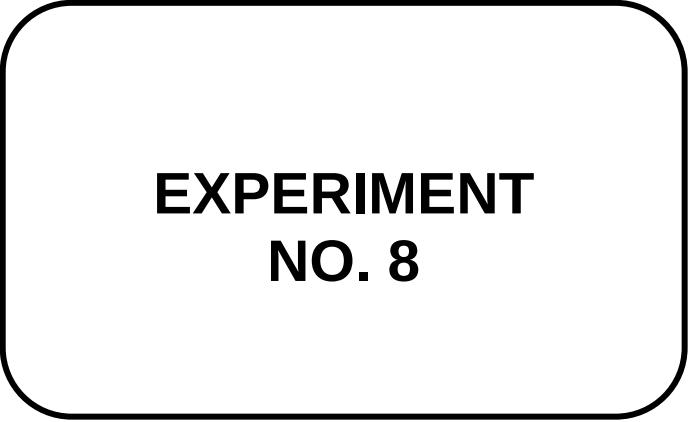
Step 12: Attempt to create a file.

```
jenny@ubuntu:/home/joe$ touch jenny.txt
```

```
jenny@ubuntu:/home/joe$ touch jenny.txt
touch: cannot touch 'jenny.txt': Permission denied
jenny@ubuntu:/home/joe$
```

Notice we do not have permission to create the file.

Step 13: Close all remaining windows.



EXPERIMENT NO. 8

EXPERIMENT NO.8

NAME: -

Remote Access

OBJECTIVES: -

Establish a remote connection to Ubuntu virtual machine from your host computer through Telnet.

BACKGROUND/ SCENARIO: -

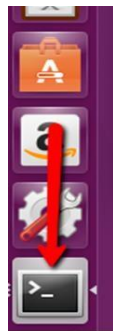
You will use Telnet to establish remote connections to a host. S Telnet is an insecure method for accessing a Telnet host.

REQUIRED RESOURCES: -

- PC with Ubuntu 22.04 Desktop LTS installed in a VirtualBox or VMware virtual machine.

Open a terminal window in Ubuntu

Click on the terminal icon to open a terminal window.



Step 2: Install Telnet by CLI

- First you need to install the Telnet service by command line interface in Ubuntu desktop.
- Update the newly install packages on your operating system by giving the command **sudo apt-get update**.

```
akademika@akademika-VirtualBox:~$ sudo apt update
[sudo] password for akademika:
Hit:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
46 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Once its read all your packages your are ready to install the Telnet.

For installing Telnet service give the command **sudo apt-get install telnetd** at the command prompt.

```
akademika@akademika-VirtualBox:~$ sudo apt install telnetd
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

Type y to continue the installation process. Wait for a while.

Once the installation process is completed you can establish remote connections to a host through telnet.

Telnet to localhost

At the command prompt, enter the following command:

C:\Users\Dell>**telnet and the IP address of your Ubuntu virtual machine.**

You are prompted for a login account and password for an account that exists on the host:
Ubuntu

login: **Akademika**

Password: **admin**

At the command prompt, enter the following command to exit this Telnet session:

C:\Users\Dell>ping 192.168.1.10

Ping statistics with the IP address of your Ubuntu virtual machine.

```
Microsoft Windows [Version 10.0.22631.3374]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Dell>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:
Reply from 192.168.1.10: bytes=32 time=18ms TTL=64
Reply from 192.168.1.10: bytes=32 time=2ms TTL=64
Reply from 192.168.1.10: bytes=32 time=1ms TTL=64
Reply from 192.168.1.10: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 18ms, Average = 5ms
```

Access the remote connection

C: \Users\Dell> telnet 192.168.1.10

```
Telnet 192.168.1.10
Ubuntu 22.04.4 LTS
akademika-VirtualBox login: akademika
Password:
Welcome to Ubuntu 22.04.4 LTS (GNU/Linux 6.5.0-25-generic x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

52 updates can be applied immediately.
25 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

7 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Mon Apr  1 16:30:56 IST 2024 from 192.168.1.9 on pts/1
akademika@akademika-VirtualBox:~$
```

You have successfully logged into your Ubuntu virtual machine through Telnet.

Step 5: Verify the connectivity

At the command prompt, type `sudo apt update`
akademika@akademika-VirtualBox:~\$ sudo
apt update

```
akademika@akademika-VirtualBox:~$ sudo apt update
[sudo] password for akademika:
Hit:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
46 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Now, at the command prompt type `sudo apt install nmap`
akademika@akademika-VirtualBox:~\$ sudo apt
install nmap

```
akademika@akademika-VirtualBox:~$ sudo apt install nmap
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nmap is already the newest version (7.91+dfsg1+really7.80+dfsg1-2ubuntu0.1).
0 upgraded, 0 newly installed, 0 to remove and 46 not upgraded.
```

Exit the remote host connection by typing the command `exit` or `logout`.

Accessing a Remote Host

Pick a partner and change the network adapters on both of your virtual machines to bridged. To do this you will need to release your cursor from the virtual machine, go to **Machine > Settings** and click **Network** and change **Attached to: NAT** to **Attached to: Bridged Adapter**. Wait for the network to reconnect. Now check your IP address by typing in the following command:

akademika@akademika-VirtualBox:~\$ **ip addr show**

```
akademika@akademika-VirtualBox:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:eb:4a:30 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.10/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
        valid_lft 84507sec preferred_lft 84507sec
    inet6 fe80::86f1:53fe:3633:d718/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Note: If you did not receive a new IP address, click the network icon (). Disconnect and reconnect the wired connection.

Repeat the Telnet commands but this time instead of localhost use your partner's IP

**EXPERIMENT
NO. 9**

EXPERIMENT NO.09

NAME: -

Hardening a Linux System Objectives

OBJECTIVE: -

Demonstrate the use of a security auditing tool to harden a Linux system.

BACKGROUND/SCENARIO: -

Auditing a system for potential misconfigurations or unprotected services is an important aspect of system hardening.

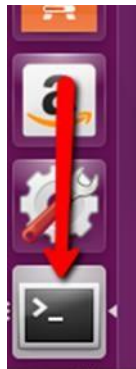
Lynis is an open source security auditing tool with an automated set of scripts developed to test a Linux system.

REQUIRED RESOURCE: -

- PC with Ubuntu 22.04 Desktop LTS installed in a VirtualBox or VMware virtual machine.

Step 1 : Open a terminal window in Ubuntu.

- a) Click the terminal icon to open a terminal window.



Step 2: Install the Lynis Auditing Tool.

- a. At the command prompt, enter the following command

```
akademika@akademika-VirtualBox:~$ sudo apt install lynis.
```

```
akademika@akademika-VirtualBox:~/Desktop$ sudo apt install lynis
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  menu
Suggested packages:
  apt-listbugs debsecan debsums tripwire samhain aide fail2ban menu-l10n gksu
  | kde-runtime | ktsuss
The following NEW packages will be installed:
  lynis menu
0 upgraded, 2 newly installed, 0 to remove and 105 not upgraded.
Need to get 581 kB of archives.
After this operation, 3,164 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

Type Y to continue the installation.

Step 5: Run the Tool

Click on the menu icon. Locate and then click on lynis auditing tool.

Step 4: Run the Tool

- Open the application and when prompted for password type the password for your Ubuntu virtual machine.

```
About to execute /usr/sbin/lynis audit system --no-colors.
This command needs root privileges to be executed.
Using sudo...
Enter akademika password at prompt.
[sudo] password for akademika: █
```

Note – Once you entered the password your application terminal will automatically start auditing the Ubuntu virtual machine.

As displayed above, the tool will begin auditing using the user **cisco** as the auditor. Notice: You will receive **warnings**.

- To continue with each stage of the audit press **Enter**. You will receive warnings as shown below.

```
[+] Boot and services
-----
- Service Manager [ systemd ]
- Checking UEFI boot [ DISABLED ]
- Checking presence GRUB2 [ FOUND ]
  - Checking for password protection [ WARNING ]
- Check running services (systemctl) [ DONE ]
  Result: found 23 running services
- Check enabled services at boot (systemctl) [ DONE ]
  Result: found 37 enabled services
- Check startup files (permissions) [ OK ]

[ Press [ENTER] to continue, or [CTRL]+C to stop ]
```

c. You will receive suggestions, as shown below.

```
[+] Users, Groups and Authentication
-----
- Search administrator accounts [ OK ]
- Checking for non-unique UIDs [ OK ]
- Checking consistency of group files (grpck) [ OK ]
- Checking non unique group ID's [ OK ]
- Checking non unique group names [ OK ]
- Checking password file consistency [ OK ]
- Query system users (non daemons) [ DONE ]
- Checking NIS+ authentication support [ NOT ENABLED ]
- Checking NIS authentication support [ NOT ENABLED ]
- Checking sudoers file [ FOUND ]
  - Check sudoers file permissions [ OK ]
- Checking PAM password strength tools [ SUGGESTION ]
- Checking PAM configuration files (pam.conf) [ FOUND ]
- Checking PAM configuration files (pam.d) [ FOUND ]
- Checking PAM modules [ FOUND ]
- Checking LDAP module in PAM [ NOT FOUND ]
- Checking accounts without expire date [ OK ]
- Checking accounts without password [ OK ]
- Checking user password aging (minimum) [ DISABLED ]
- Checking user password aging (maximum) [ DISABLED ]
- Checking expired passwords [ OK ]
```

d. You will receive a notification for any configuration that is weak as shown below:

```
[+] Banners and identification
-----
- /etc/motd [ NOT FOUND ]
- /etc/issue [ FOUND ]
  - /etc/issue contents [ WEAK ]
- /etc/issue.net [ FOUND ]
  - /etc/issue.net contents [ WEAK ]

[ Press [ENTER] to continue, or [CTRL]+C to stop ]
```

e. You will receive detailed security enhancement suggestions as well as a final summary which provides the location where you can find the log file.

```

Lynis security scan details:

Hardening index : 56 [#####          ]
Tests performed : 188
Plugins enabled : 0

Quick overview:
- Firewall [X] - Malware scanner [X]

Lynis Modules:
- Compliance Status [NA]
- Security Audit [V]
- Vulnerability Scan [V]

Files:
- Test and debug information : /var/log/lynis.log
- Report data : /var/log/lynis-report.dat

```

Step 3: Review Results.

a. Scroll up to the results section after the tool is finished running.
 How many Warnings did you receive? _____ Answers
 will vary. There was 1 warning in this example.
 How many Suggestions did you receive? _____ Answers
 will vary. There were 33 suggestions in this example.

b. Scroll through the suggestions and select one. You will research a suggestion that
 you can implement to address the issue.
 Which suggestion are you addressing?

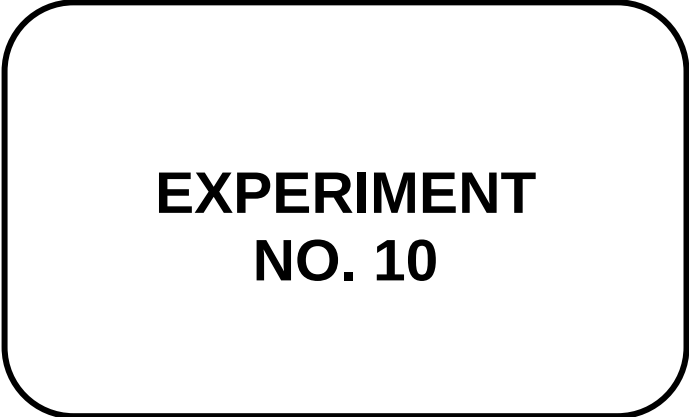
Answers will vary.

What is your suggested solution?

Answers will vary

References:-

Lynis: <https://cisofy.com/lynis/>



EXPERIMENT NO. 10

EXPERIMENT NO.10:-

NAME: -

Using Steganography.

OBJECTIVE: -

Use steganography to hide a document within a JPEG file.

BACKGROUND/ SCENARIO: -

Steghide is an open source steganography program that hides data in various types of files such as audio and image files. You are going to hide a data file within an image file.

REQUIRED RESOURCE: -

- PC with Ubuntu 22.04 Desktop LTS installed in a VirtualBox or VMware virtual machine

Step 1: Open a terminal window in Ubuntu.

- a) Click on the terminal icon to open a terminal.



Step 2: Install Steghide on your operating system

Step 3: Commands for installing Steghide

- a) First update your Ubuntu for locating the files by entering the command `sudo apt update`.

```
akademika@akademika-VirtualBox:~/Desktop$ sudo apt update
Get:1 http://security.ubuntu.com/ubuntu jammy-security InRelease [110 kB]
Hit:2 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Get:3 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease [119 kB]
Hit:4 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease
Fetched 229 kB in 3s (76.6 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
105 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

- b) Now, install steghide software by entering the command `sudo apt install steghide`.

```
akademika@akademika-VirtualBox:~/Desktop$ sudo apt install steghide
[sudo] password for akademika:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libmcrypt4
Suggested packages:
  libmcrypt-dev mcrypt
The following NEW packages will be installed:
  libmcrypt4 steghide
0 upgraded, 2 newly installed, 0 to remove and 105 not upgraded.
Need to get 213 kB of archives.
After this operation, 701 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

- c) Type Y to continue.

Step 4: Run Steghide.

- Open your Browser and search for High resolution images (HD images)
- Download any one image
- Name the image as clock.jpg.
- Open Libre office and create a word document with some secret text that you want to embed
- Name the word document as secret.odt
- Note: the size of the image should be greater than the size of the secret document.

```
akademika@akademika-VirtualBox:~/Downloads$
libreoffice secret123.odt &What is the message in the
secret.odt?
```

- g) Close the **secret.odt** file when done.
- h) Download the GNU image manipulation program
- i) Open the terminal and type `sudo apt install gimp`
- j) Enter **gimp clock.jpg &** at the prompt to view the image file

akademika@akademika-VirtualBox:~**gimp clock.jpg &**

```
akademika@akademika-VirtualBox:~/Downloads$ gimp clock.jpg
Gtk-Messsage: 14:35:21.347: Failed to load module "canberra-gtk-module"
Gtk-Messsage: 14:35:36.927: Failed to load module "canberra-gtk-module"
gimp_color_transform_new: using babl for 'sRGB IEC61966-2.1' -> 'GIMP built-in sRGB'
gimp_color_transform_new: using babl for 'sRGB IEC61966-2.1' -> 'GIMP built-in sRGB'
void gimp_gegl_convert_color_profile(GeglBuffer*, const GeglRectangle*, GimpColorProfile*, GeglBuffer*, const GeglRectangle*, GimpColorProfile*, GimpColorRenderingIntent, gboolean, GimpProgress*): converting buffer took 0.0274 seconds
```

- k) Close the **clock.jpg** file when done.
- l) Change the directory according to the location of your image and secret document
- m) The image and secret word document must be in the same directory.
- n) At the command prompt, enter the following command :

akademika@akademika-VirtualBox:~/Downloads\$ **steghide embed -cf clock.jpg -ef secret123.odt**

This command takes the jpeg file called “clock.jpg” and uses it as a carrier to embed the document, **secret123.odt**, into it.

- o) When prompted for a passphrase, use **Cisco**. Re-enter the passphrase when prompted.

```
akademika@akademika-VirtualBox:~/Downloads$ steghide embed -cf clock.jpg -ef secret12345.odt
Enter passphrase:
Re-Enter passphrase:
embedding "secret12345.odt" in "clock.jpg"... done
```

- p) You have embedded the document, **secret123.odt**, into the image file, clock.jpg.
- q) Open the files, **secret123.odt** and **clock.jpg**. Did these files change? ____

No. The files did not change.

Step:5 Verify the hidden file.

- r) Type the following command in terminal.

akademika@akademika-VirtualBox:~/Downloads\$ **steghide info clock.jpg**

```
akademika@akademika-VirtualBox:~/Downloads$ steghide info clock.jpg
"clock.jpg":
  format: jpeg
  capacity: 250.6 KB
Try to get information about embedded data ? (y/n) y
Enter passphrase:
  embedded file "secret12345.odt":
    size: 9.3 KB
    encrypted: rijndael-128, cbc
    compressed: yes
```

- s) Type **y** at the prompt. (Do not press **Enter**).
- t) Enter the passphrase **Cisco** and press **Enter**.
- u) The results below shows that the file, secret123.odt, is encrypted and compressed.

Step 3: Extract the hidden file.

- a) Type the following command in terminal.

akademika@akademika-VirtualBox:~/Downloads\$ **steghide extract -sf clock.jpg**

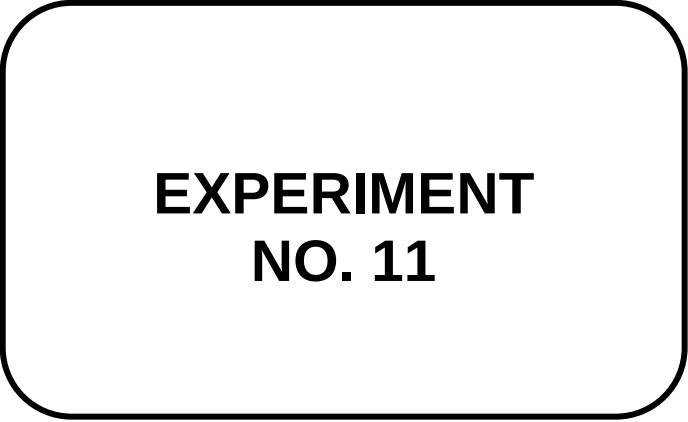
```
akademika@akademika-VirtualBox:~/Downloads$ steghide extract -sf clock.jpg
Enter passphrase:
the file "secret12345.odt" does already exist. overwrite ? (y/n) y
wrote extracted data to "secret12345.odt".
```

- b) Enter the passphrase, **Cisco**, and press **Enter**.
- c) Enter **y** when prompted to overwrite the existing **secret123.odt** file with the new extracted **secret123.odt** file.
- d) You have extracted the file. Open the extracted **secret.odt** file with LibreOffice.
- e) Could you open the file? Is the secret message the same as before?

The file can be opened and the message is the same as before.

References:-

Steghide: <http://steghide.sourceforge.net/>



EXPERIMENT NO. 11

EXPERIMENT NO. 11

NAME: -

Detecting Threats and Vulnerabilities

OBJECTIVE: -

Use Nmap, a port scanner and network mapping tool to detect threats and vulnerabilities on a system.

BACKGROUND/ SCENARIO: -

Network Mapper, or Nmap, is an open source utility used for network discovery and security auditing. Administrators also use Nmap for monitoring hosts or managing service upgrade schedules. Nmap determines what hosts are available on a network, what services are running, what operating systems are running, and what packet filters or firewalls are running.

REQUIRED RESOURCES: -

- PC with Ubuntu 22.04. LTS installed in a virtual machine - you can use the VM from labs completed in chapter 2.

Open a terminal window in Ubuntu.

Click on the **terminal** icon to open a terminal.



Install Nmap on your Ubuntu Operating System

Open in terminal, type the command **sudo apt install nmap**

```
akademika@akademika-VirtualBox:~$ sudo apt install nmap
[sudo] password for akademika:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

Run Nmap.

How To Use Nmap

Nmap is pretty easy to use if you're familiar with command-line interfaces. As it's already installed on most Linux/Unix-based distributions, **you just have to execute the 'nmap' command from any terminal**, and that's it. It will display several options for you.

Advanced users will also be able to use Nmap along with other system scripts and automated tasks in order to maximize the powers of this tool.

Nmap port scan command

One of the most basic Nmap commands for a scan is the nmap port scan command:

That's how you use Nmap.

Nmap Command Examples - Full Tutorial

Let's get to know a few useful command-line based best Nmap scans that can be performed.

1. Scan a Range of IP Address: To scan a range of IP addresses, the Nmap command is as follows:

Helps you find active IP addresses within specified range.

```
nmap 192.168.1.1-24
```

2. Port Scanning: A technique that uses an application called port scanner to identify open ports in computer system. Both cybercriminals and security professionals use port scanning to check vulnerabilities in a network. There are multiple commands in Nmap for scanning ports such as:

To scan TCP port 80, the following Nmap command can be used:

```
nmap -p T:80 192.168.1.1
```

To scan UDP port 53:

```
nmap -p U:53 192.168.1.1
```

To scan the range of ports:

```
nmap -p 80-160 192.168.1.1
```

Note: Open ports refers to a TCP or UDP port number that is configured to accept packets. In contrast, a port that reject connections or ignores all packets is a closed ports.

We can also combine all these commands to scan multiple ports:

```
nmap -p U:53, 112, 135, T:80, 8080 192.168.1.1
```

3. Ping Scan Using Nmap: It can be used for host discovery and the following command can be used:

```
nmap -sP 192.168.1.1/20
```

4. Most Popular Ports Scanning: The most popular TCP ports can be scanned using TCP SYN scan and the following command exists for this purpose:

It runs by default when running nmap as root administrator. It is the most popular scan option according to nmap.org, quick and efficient, this scan can indicate open, filtered closed ports states. Also Known as half-open scan, it never completes the full TCP connection, so is less likely to be blocked by firewalls.

```
nmap -sS 192.168.1.1
```

the above command is used for stealthy scan

For OS fingerprinting, the following command can be used:

```
nmap -sT 192.168.1.1
```

5. Display Open Ports: The command for displaying open ports on the network is as follows:

```
nmap --open 192.168.1.1
```

```
nmap --open server2.gl.biz
```

```
nmap --open 192.168.0.1
```

6. Exclude Host/ IP Addresses for the Scan: In order to exclude the hosts from the Nmap scan, you can use the following Nmap commands:

If you are scanning a number of hosts/networks, then you can exclude hosts/IPs from a scan by using:

```
nmap 192.168.1.1-24 --exclude 192.168.1.4
```

or

`nmap 192.168.1.1-24 --exclude 192.168.1.3, 192.168.1.7`

for excluding more than one host.

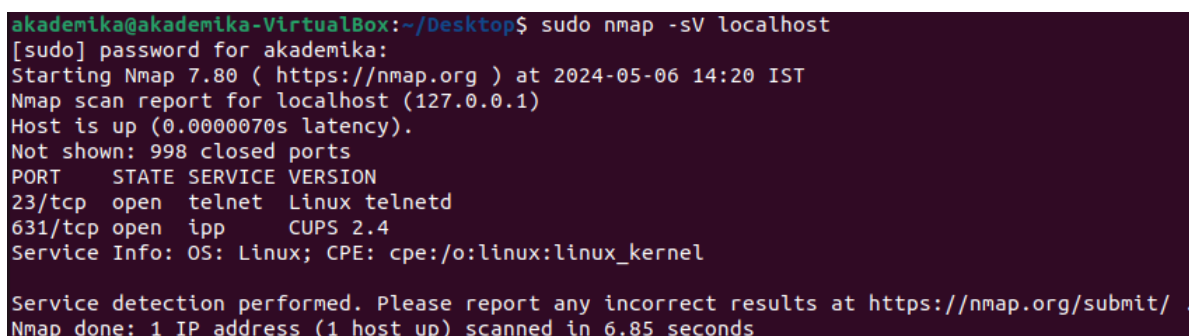
7. Scan + OS and service detection with fast execution

Using the “-A” parameter enables you to perform OS and service detection, and at the same time we are combining this with “-T4” for faster execution.

8. Detect service/daemon versions

This can be done by using -sV parameters

`nmap -sV localhost`



```
akademika@akademika-VirtualBox:~/Desktop$ sudo nmap -sV localhost
[sudo] password for akademika:
Starting Nmap 7.80 ( https://nmap.org ) at 2024-05-06 14:20 IST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000070s latency).
Not shown: 998 closed ports
PORT      STATE SERVICE VERSION
23/tcp    open  telnet  Linux telnetd
631/tcp    open  ipp     CUPS 2.4
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.85 seconds
```

As you can see here:

```
akademika@akademika-VirtualBox:~$ nmap -sV localhost
Starting Nmap 7.60 at 2018-10-01 09:28 -03
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000020s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 997 closed ports
PORT      STATE SERVICE VERSION
111/tcp    open  rpcbind 2-4 (RPC #100000)
631/tcp    open  ipp     CUPS 2.2
902/tcp    open  ssl/vmware-auth VMware Authentication Daemon 1.10 (Uses VNC, SOAP)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.96 seconds
```

9. Scan using TCP or UDP protocols

One of the things we love most about Nmap is the fact that it works for both TCP and UDP protocols. And while most services run on TCP, you can also get a great advantage by scanning UDP-based services. Let's see some examples.

Standard TCP scanning output:

```
akademika@akademika-VirtualBox:~/Desktop$ sudo nmap -sU localhost
Starting Nmap 7.80 ( https://nmap.org ) at 2024-05-06 14:20 IST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000039s latency).
Not shown: 998 closed ports
PORT      STATE      SERVICE
631/udp    open|filtered ipp
5353/udp    open|filtered zeroconf

Nmap done: 1 IP address (1 host up) scanned in 1.34 seconds
```

UDP scanning results using “-sU” parameter:

```
akademika@akademika-VirtualBox:~$ nmap -sU localhost
Starting Nmap 7.60 at 2018-10-01 09:37 -03
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000021s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 997 closed ports
PORT STATE SERVICE
68/udp open|filtered dhcpd
111/udp open rpcbind
5353/udp open|filtered zeroconf
```

Capture SSH keys.

Type the following command in the terminal to initiate a script scan:

akademika@akademika-VirtualBox:~\$ **nmap -A localhost**

```
akademika@akademika-VirtualBox:~/Desktop$ sudo nmap -A localhost
Starting Nmap 7.80 ( https://nmap.org ) at 2024-05-06 14:21 IST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000039s latency).
Not shown: 998 closed ports
PORT      STATE SERVICE VERSION
23/tcp    open  telnet  Linux telnetd
631/tcp    open  ipp     CUPS 2.4
|_ http-robots.txt: 1 disallowed entry
|_/
|_ http-server-header: CUPS/2.4 IPP/2.1
|_ http-title: Home - CUPS 2.4.1
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6.32
OS details: Linux 2.6.32
Network Distance: 0 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 15.57 seconds
```

You captured the SSH keys for the host system. The command runs a set of scripts built

References: -

Nmap: <https://nmap.org/>

**EXPERIMENT
NO. 12**

EXPERIMENT NO. 12

NAME: -

Lab – Threat Identification.

OBJECTIVES: -

Explore the security features used by organizations to keep data safe.

Part 1: Exploring the Threat of Cyberattacks.

Part 2: CIA Triad.

Background / Scenario

The threats posed by the cyber world are real. These threats have the potential to wreak havoc on life in a computer centric world. Understanding these threats is important to everyone and in order to combat them, the world needs committed individuals that can recognize threats, and outmaneuver and outsmart cyber criminals. In order to develop the talent needed, organizations like CompTIA, Cisco Systems and ISC2 have created programs to educate and certify cyber professionals.

Required Resources

- PC or mobile device with Internet access

Exploring the Threat of Cyberattacks

Cyberattacks top the list of threats facing countries around the world. When people think of threats to national or world security, most people think of physical attacks or weapons of mass destruction. The fact is cyber threats top the list in over twenty countries around the world. The ranking of cyberattacks in the number one spot reveals a few things about how society has changed. Computers and computer networks affect the way we learn, shop, communicate, travel, and live. Computer systems control almost every aspect of our lives. The disruption of computer systems and computer networks can have a devastating impact on modern life. Electrical power generation and distribution systems, water treatment and supply systems, transportation, and financial systems are all targets of cyberattacks. Each of these systems has been a victim of cyberattacks. Watch the video below. Break into groups of 3-4 people. After viewing the video, answer the questions below.

Research Threats.

In step 1, you will research threats.

Click <https://www.youtube.com/watch?v=WnUuLzXCIEk> to view the video. According to the video, what is the most dangerous weapon in the world? Why? Do you agree?

A computer in the wrong hands is the most dangerous weapon in the world. Other weapons have the ability to create great harm, but computers are present everywhere and people can use them to attack one another or large organizations, in anonymity. They can also be used to bully a classmate, or steal someone's identity.

List five ways a cyber-criminal can use computers to break the law. Can any of the crimes you listed affect you personally? Have you or your family members been affected by these crimes?

A computer in the wrong hands can be used to steal someone's identity, make illegal purchases using other people's credit cards, or posting objectionable pictures on the web.

Answers will vary.

Have any of the potential threats portrayed in the video actually happened? Click <https://www.nytimes.com/subscription?campaignId=4YHUUW> to learn more about these attacks.

Answers will vary.

Explore Recent Attacks.

The impact and scope of recent cyberattacks have many business and government officials concerned. Click <https://www.information-age.com/top-10-most-devastating-cyber-hacks-2015-123460657/>

to review the top 10 most devastating cyber hacks or 2015.

How many people did the US Office of Personnel Management data breach affect?

This breach was one of the biggest ever of a US government system. The data theft consisted of stealing addresses, health information, and financial details of 19.7 million people who were subjected to government background checks, as well as 1.8 million others.

Describe the TalkTalk attack of 2015. Who was responsible and what did the cyber criminals steal?

The mobile phone provider was the target of a group of teenage hackers who stole the customer information details of over 20,000 customers.

CIA Triad

Confidentiality, integrity, and availability are the three fundamental cybersecurity principles. These three principles make up the CIA triad. The elements of the triad are the three most crucial components of security. All cybersecurity professionals should be familiar with these core principles.

Explore the CIA Triad.

Click <https://www.youtube.com/watch?v=bhLbnOa4wno> to view the video. What is data confidentiality? Why is data confidentiality so important to people and organizations?

Confidentiality is protecting the privacy of information. People and organizations use information systems for many day-to-day activities. When using computer systems, individuals and business are often required to share very sensitive data. The organizations that collect this data are responsible for its safety and privacy. In the wrong hands, sensitive information can be misused to commit illegal activity.

What is data integrity? Name three ways data integrity or trustworthiness is affected.

Data integrity is the principle of protecting and maintaining the consistency, accuracy, and trustworthiness of data over its entire life cycle. Equipment failure, errors, mistakes by people, or attacks by cyber-criminal who intentionally change, delete or damage data, are all examples of how data integrity is affected.

What is system availability? What can happen if a critical computer system is no longer available?

The cybersecurity principle of Availability states that information and information systems must be available when needed. Availability systems must have properly functioning computing systems, services and communication channels. Loss of availability can affect any services and access to data on the systems. In the case of a bank system availability failure, a bank customer would not be able to deposit or withdraw money to or from the bank.

Explore Cyberattacks.

Click <https://youtu.be/5qKnZcQQJ4w> to watch a video. What were the cybercriminals trying to do? What time of the day did the attack occur? Are network attacks likely to occur after hours? Why?

The cybercriminals were attempting to gain access to the Garrison Inc. network. The attack took place at 17:00 on a Friday. Most external network attacks will occur after work hours because the organization is less likely to detect the attack.

